

Limiting For-Profit Provision in Nursing Home Markets

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Abstract

We examine whether policies that ban for-profit providers and allow for subsequent takeovers by not-for-profits can be effective at addressing quality shortfalls. We consider the US nursing home industry, where quality provision has been a concern for many decades. Our motivating evidence suggests that not-for-profit providers choose higher-quality inputs than their for-profit counterparts but are more prevalent in higher socioeconomic markets and serve wealthier residents. Thus, for-profit providers play an important role in providing access. To explain these facts and explore counterfactual policies, we estimate a structural model of nursing home demand and supply that allows firms to have nonpecuniary motives and costs that differ across provider types. The structural model reveals that for-profit providers have a strong cost advantage for serving needier residents. This results in them choosing a lower quality and lower priced product to maximise their margins. Not-for-profit providers choose higher quality, and therefore higher prices, because they prefer to avoid directly competing with for-profit providers by serving residents with stronger preferences for quality, and to a lesser extent their non-pecuniary motive. Therefore, banning for-profit providers while allowing for takeovers reduces consumer surplus because not-for-profits that take over for-profits significantly raise prices to cover their higher costs. These results underscore the role that for-profit providers play in expanding access to nursing home care.

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1 Introduction

In health and education settings, policymakers and consumers frequently complain that some firms provide low-quality services. Two common features in these markets are the ability of policymakers to decide who is allowed to compete through certification or entry requirements and competition between for-profit (FP) and not-for-profit (NFP) providers. Policymakers and other commentators often blame FP providers for quality shortfalls, which leads these individuals to question whether FP providers should be allowed to compete. For example, policymakers in California sought to limit the operations of for-profit colleges due to systemic quality issues (Mello, 2019).¹ Changing the composition of firm types may not be the most obvious policy solution to quality shortfalls, but in practice, many other reforms have had limited success.² Thus, in this paper, we ask whether policymakers should increase NFP provision and decrease FP provision in these kinds of markets.

We study this question in the context of the US nursing homes industry, where low quality provision is a severe problem that policymakers have long struggled to solve.³ In recent years, New York policymakers considered banning all new FP providers (Hogan, 2021). Ultimately, this did not pass, but several states including Massachusetts, New Jersey, and New York imposed strict profit caps of 5% of expenses on for-profit nursing homes (Jaffe and News, 2021).

In this paper, we study the effects of these types of policies by studying the effects of bans on FP nursing homes, allowing for subsequent takeovers by NFP nursing homes. If NFP providers have nonpecuniary motives, then their incentives will be more in line with a social planner's. Increased NFP provision could therefore lead to increased quality. In addition, Weisbrod (1988) suggests that NFP firms can solve information asymmetries. When quality is difficult to observe, consumers may have an explicit preference for NFP providers, reflecting trust in their nonprofit objectives.

On the other hand, if the business models of FP and NFP providers systematically differ, then so could their costs. If FP providers are more cost-efficient, then they could play an important role

¹Similar concerns have arisen in for-profit medical schools and nursing homes (Knight, 2021; Campbell and Harrington, 2024; Penzenstadler, 2022).

²One example is pay-for-performance where the overall results are mixed in both health and education settings (Cohodes et al., 2023; Fryer, 2013; Gupta, 2021; Alexander, 2020; Finkelstein et al., 2018). Another is the increased provision of information about quality – ‘report cards’ – that also presents mixed results in health and education settings (Dranove, 2011; Dranove and Jin, 2010). The key issue is that firms can game payment schemes and report cards to avoid improving quality.

³A study report by the National Academies of Sciences, Engineering, and Medicine (2022) provides a summary of the evolution of policy and reforms to improve nursing home quality going back several decades. In their introduction, they reflect on their predecessor report from 1986, which raised serious concerns regarding quality provision. They also highlight how COVID-19 laid bare the continued quality failings in nursing homes. Approximately 40% of all US COVID-19 deaths were among nursing home residents (SteelFisher et al., 2021).

in expanding access to nursing home care, serving at lower price and serving residents who may not otherwise have received access to nursing home care. In addition, FP providers serve roughly two-thirds of the market. A ban on all FP providers would dramatically reduce the supply of services and increase the distance between a resident's prior residence and nursing home, which may be undesirable if residents value proximity to service.⁴ Summarising, the effects of these policies are ambiguous and we estimate a structural model of nursing home demand and supply, encompassing these forces, to examine their effects on quality, access and welfare.

We begin by using a nationwide sample of nursing homes to document patterns of quality and access across FP and NFP providers. In this paper, the main characterisation of socioeconomic background will be whether the resident is a private payer, who pays the private price set by the nursing home out of pocket, or is covered by Medicaid, which typically requires exhaustion of wealth and assets for eligibility.⁵ We use cross-sectional variation as FP and NFP providers operate distinct business models and rarely change their for-profit status from year-to-year.

We find that NFP providers choose higher quality – measured in terms of higher nursing intensity – than their FP counterparts. On the other hand, FP providers are more prevalent in poorer markets and serve more Medicaid residents than NFP providers. This holds even within markets where NFP and FP nursing homes compete. The result on quality is consistent with the literature, but our result regarding NFP provision across and within markets is new and may be surprising given the supposed nonpecuniary motives of a NFP provider.

The key to understanding these results is to conceptualise NFP providers as offering a higher-quality, and thus higher-priced product, which is preferred by residents who value quality more, especially private-payers. FP providers provide a lower-priced, and thus lower-quality product that is suited to residents who value quality less, including Medicaid payers who may pay the private price for some part of their stay.⁶ These results underscore that the welfare effects of decreasing FP provision and increasing NFP provision are ambiguous. More NFP provision might increase quality on average, but access to nursing home care could be reduced, especially for Medicaid residents.

The reduced-form facts alone are not sufficient to justify our interpretation of the observed quality access patterns. In particular, we do not know why NFP providers choose higher quality than their FP counterparts. For instance, is it driven by differing preferences for quality across residents, the nonpecuniary motives of NFP firms, or different cost functions? In addition, our

⁴Most papers estimating resident demand in health economics include distance in utility and find that it is a significant determinant of resident choice (Hackmann, 2019; Ho, 2006; Town and Vistnes, 2001). A similar treatment of distance can also be found in education settings (Dinerstein et al., 2020).

⁵We introduce payer types in greater detail, including Medicare payers, when we discuss the setting in Section 2.

⁶Since private prices come from state cost reports, this result is based on California data (where our structural estimation takes place).

reduced-form facts do not indicate whether FP providers prefer serving Medicaid residents and whether NFP providers prefer serving private-payers. This depends on the relative margins across firm and payer-types, which in turn depend on demand- and supply-side factors. Marginal costs are not directly observed in the data. Therefore, to complete these explanations, we need a structural model of nursing home demand and supply that generates estimates of all the parameters of interest. In addition, for the counterfactuals of interest, there are no natural experiments that provide exogenous variation in the number of FP versus NFP providers. Thus, a structural model of nursing home demand and supply is needed to understand the welfare effects of novel regulations.

We estimate a structural model of nursing home demand and supply on markets in California following Hackmann (2019). Nursing home demand is modelled as a standard discrete choice problem. We observe many individual characteristics which allows us to model price and quality elasticities that vary significantly at the individual level. This gives rise to the potential for market-segmentation from demand-side factors. On the supply side, nursing homes compete in an oligopoly game by choosing qualities and private prices to maximise an objective that includes profits and nonpecuniary motives towards quantity of care. Following our motivating evidence, we consider a supply side that is significantly richer than other papers in this literature. We allow cost functions and the quantity-of-care motive to vary systematically across NFP and FP providers. Cost functions also vary systematically with the payer-type quantities served, reflecting that the cost of serving patients of different payer-types varies due to their differing care needs. This allows us to capture payer-type margins that vary at the firm level and will be the key to explaining the patterns of access and provision.

Our model estimates reveal that NFP providers have a nonpecuniary value for providing an additional bed-day of care of approximately \$28 – 15% of their private price or one-third of their average margin. The corresponding value for FP providers is not significantly different from zero. On the cost side, we find that overall, NFP providers have higher quality costs. Our quality measure – nursing intensity – aggregates across two different types of nurses. NFPs use a mix of nurses which includes more of the higher qualification and higher cost nurses, which explains quality cost differences. Regarding payer-type quantities, we find that FP providers have significantly lower payer-type cost parameters than NFP providers. Some of these cost differences are likely attributable to differences in unobserved quality. These include dimensions of quality related to accommodation quality and other aspects of nursing care not captured by the nursing hours we include, such as differences in the usage of nursing assistants. For example, for Medicaid payers where the cost difference is around \$45 per bed-day on average, around 40% of the cost difference is attributable to differences in unobserved quality. The remainder we interpret as differences in the efficiency of providing nursing home care.

We then use the model to explain the in-sample patterns of quality, access and provision. We

find that NFP providers choose higher quality both because of their nonpecuniary motive and also a market segmentation strategy that serves residents who value quality more, including private-payers, with the latter having a stronger effect. Next, turning to patterns related to access and provision, we find that FP providers make significantly larger margins on Medicaid payers than do NFP providers. This explains why FP providers are more prevalent in markets with lower socioeconomic status, which as we showed have more Medicaid residents. NFP providers cannot compete as well with the lower costs of FP providers and ultimately serve markets with higher socioeconomic status and fewer Medicaid residents.

Finally, we turn to our policy counterfactuals, where we ban FP nursing homes and have them be taken over by a local NFP nursing home. Since government providers are included in our pool of NFP providers, this could also be thought of as a government takeover.⁷ The new NFP nursing home then follows a NFP business model, although nursing hours are still endogenous. That is, we replace all the FP cost parameters with their NFP counterparts and set the unobserved cost shocks to the mean of the NFP unobserved cost shocks. On the demand side, since we estimate a preference for NFP providers, we also add that to a resident's evaluation of the nursing home. In these counterfactuals, we ignore any transition costs and compare the Nash-Bertrand equilibrium observed in the data prior to the change with the Nash-Bertrand equilibria resulting from the changes. We find that bans with takeovers reduce consumer surplus by \$3.7 million on average per year in the scenarios considered. This arises primarily because the NFP provider which takes over the FP provider has to significantly raise private prices and reduce quality to reduce costs as their business model is less suited to serving Medicaid residents which tend to make up a relatively larger proportion of residents served in FP nursing homes. These results suggest that reducing FP provision and increasing NFP provision are unlikely to be successful policies. They also underscore the interpretation that FP providers expand access to nursing home care by providing the market with a product that is better suited to serving residents in markets with lower socioeconomic status.

Related literature

While we combine and draw on tools from several papers and bodies of literature for our model, we view our contribution as primarily empirical. Our paper contributes to two bodies of literature. First, this project adds to the extensive literature on NFP providers by estimating a structural model of NFP and FP providers. One focus of the literature has been on understanding the differences between FP and NFP providers in various regulated markets, such as the hospital industry (Newhouse, 1970; Norton and Staiger, 1994; Sloan, 2000; Ballou and Weisbrod, 2003; Horwitz and

⁷ Around 20% of NFP providers in our pool are government providers.

Nichols, 2009; Tom and Jacobson, 2010; Bayandir, 2012; Dranove et al., 2017; Garthwaite et al., 2018), nursing home industry (Chou, 2002; Grabowski and Hirth, 2003; Ballou, 2005, 2008; Grabowski et al., 2013; Lu and Lu, 2021), and postsecondary education industry (Deming et al., 2012). However, these papers usually approach these differences from a reduced-form perspective, focusing on identifying differences in observable outcomes. For the nursing home industry, earlier works typically find that NFP providers exhibit some form of altruistic motive, typically through better-quality care in some form. However, as Weisbrod (1988) discusses, in general, NFP providers are not well understood.

Reduced-form findings of better-quality care alone are not informative about what is different about NFP providers, especially in an equilibrium with competing FP providers. A structural model allows us to consider a broad range of factors in a coherent way. Our findings reveal that while not-for-profits do have nonprofit motives, they provide higher quality also because of a market segmentation strategy. Next, we also document novel patterns of access and market segmentation and provide a complete explanation through the lens of the spatial distribution of payer-types and suitability of firm types for particular payer-types. We believe that these results on FP and NFP have some generalisability. For example, similar patterns of quality and access are seen in the postsecondary education industry (Deming et al., 2012).

Second, this paper contributes to the literature on long-term care in the US (Cheng, 2023; Ching et al., 2015; Einav et al., 2022; Gandhi, 2023; Gandhi and Olenski, 2024; Hackmann et al., 2024; Harrington et al., 2017; Lin, 2015; Olenski, 2022). A number of papers study determinants of quality and find that Medicaid reimbursement rates (Nyman, 1985; Grabowski, 2001; Hackmann, 2019) and nurse staffing ratios (Harrington et al., 2000) are important. While we borrow many aspects of Hackmann's (2019) framework, his paper focuses on studying the effect of Medicaid reimbursement rates on quality. FP versus NFP provision does not play a substantial role in his paper. Other sets of papers find that FP ownership (Chou, 2002; Grabowski and Hirth, 2003) and private equity ownership (Gupta et al., 2021) result in lower quality. These results suggest that if FP providers' inferior quality comes from investing less in quality determinants, limiting FP provision and increasing NFP provision may be welfare-enhancing. However, if FP facilities provide access to care to lower socioeconomic status populations who would not have received care otherwise, reducing FP provision may leave a void in these communities. Li et al. (2015) document that racial and ethnic minorities are more likely to obtain care from facilities with FP ownership. Our project is the first to document and explain the quality access trade-off of NFP and FP provision in a unified framework.

2 Setting

2.1 Nursing Homes

We examine Medicare-certified Skilled Nursing Facilities to study the US nursing homes industry. Skilled nursing facilities primarily provide skilled nursing care and related services for residents who require medical or nursing care and are not primarily for the care and treatment of mental diseases or personal non-nursing care (42 U.S. Code § 1395i-3). Specifically, nursing homes provide two services. First, they provide medical care to individuals who require short-term rehabilitative stays after hospitalization. Second, they provide long-term care services to individuals who require daily nursing care and additional therapy care.

The two distinct functions of the nursing home give rise to significant variation in resident characteristics, some of which we summarise in table 13 in appendix A.1. Here, we highlight the large variation in the distance between a resident's prior residence and their nursing home. Figure 1 plots a histogram of the distance between a Californian resident's prior residence and their nursing home using our individual level data from 2000-2010. Much of the mass is concentrated towards the left, with approximately 10% of stays being in the same zip and the median distance being 12.8 miles. This suggests that nursing home residents have a preference for being close to their prior residence. On the other hand, there is also a long tail leading to the average distance being 36.3 miles.

Nursing homes are primarily reimbursed on a prospective price-per-day basis. Prospective means that the prices are set by the relevant authority ahead of time. Residents can be distinguished by the funding source for their stay, or their 'payer-type'. In this paper, we focus on three payer-types: Medicare, private-pay and Medicaid. These are not exhaustive, but make up 87% of stays. Medicare covers post-acute care after hospitalisation for up to 100 days. The first 20 days are fully covered, while there is a copayment for 20-100 days.⁸ After residents exhaust those 100 days, they pay the private price set by the nursing home, unless they qualify for Medicaid. As long-term care insurance take-up is relatively low (Brown and Finkelstein, 2007), much of the payment comes from out-of-pocket. Finally, if residents exhaust their financial resources and become unable to pay prices out-of-pocket, Medicaid covers their nursing home expenses.

Medicare rates are set by the Centers for Medicare & Medicaid Services and take into account factors such as geographic adjustments and expected facility population care needs. Because of the latter, Medicare tends to pay the most.⁹ Medicaid prices are set at the State level and are relatively

⁸Since the copayment is facility invariant, we ignore the role of the copayment in our analysis.

⁹For example, in California in 2018, the average Medicare, private-pay and Medicaid prices were \$307.0, \$306.1 and \$294.2 per bed-day respectively. Average prices are calculated as total revenues divided by resident-days for each payer-type. We use California data here as data about private-pay residents is only available from state cost report data and

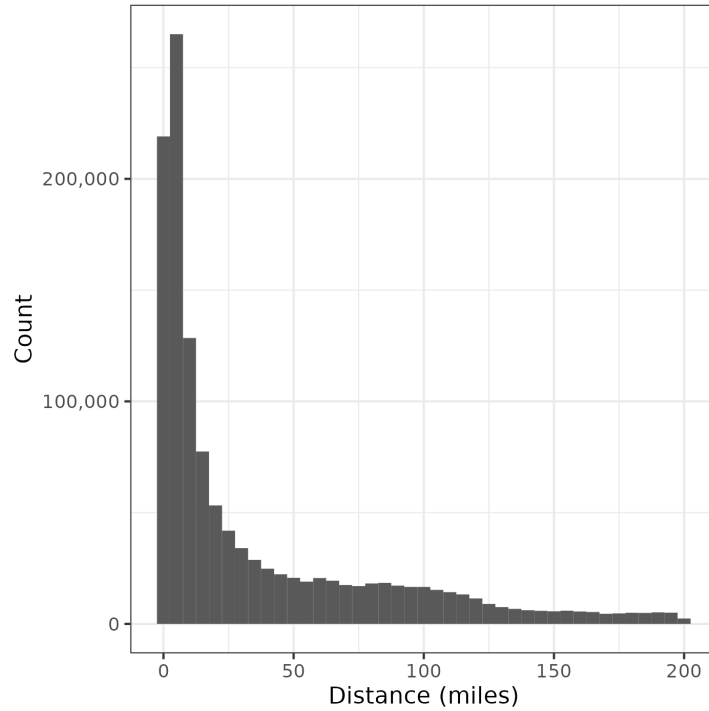


Figure 1: Histogram of distance between a resident’s prior residence and nursing home

low. Advocates and experts have recognised the low rate as a longstanding problem.¹⁰ The issue is particularly acute as Medicaid rates are not set with explicit adjustments for expected facility population care needs as with Medicare rates. Given the prospective nature of Medicaid and Medicare prices, we think of nursing homes as taking them as given.

2.2 Not-for-Profit Organisations

NFP organisations are organized and operated exclusively for specific purposes and must not inure any of the earnings to any private shareholder or individual (Internal Revenue Service, 2023).¹¹ Under section 501(c)(3) of the Internal Revenue Code, these organisations are tax-exempt from sales and property taxes and can receive tax-deductible donations.¹²

NFP organisations must file annual reports (Form 990 or its variant) which include information
our structural estimation will be based on California.

¹⁰For instance, the Committee on Improving Quality in Long-Term Care explored these concerns in its 2001 report citing policy discussion going as far back as the 1980s (Wunderlich and Kohler, 2001). One of Hackmann’s (2019) key contributions is to suggest that increasing Medicaid reimbursement rates from their relatively low rates is an effective policy for improving quality.

¹¹Specific purposes include charitable, religious, educational, scientific, literary, testing for public safety, fostering national or international amateur sports competition, and preventing cruelty to children or animals.

¹²See <https://www.irs.gov/charities-non-profits/exempt-organizations-what-are-employment-taxes>

about revenue, expenditure, and income. Also, NFP organisations may be subject to audits by the IRS or have to submit an audited financial statement depending on specific state laws.¹³ We take the effectiveness of this regulation as given and consider NFP and FP as firms of different types throughout our analysis.

2.3 NFP and FP Nursing Homes

Table 1: Summary Statistics from facility-level data

	NFP	FP
Provider count in 2000	5,889	11,075
	(33.3%)	(66.7%)
Provider count in 2018	4,463	10,437
	(30.0%)	(70.0%)
Average entries per year	34.1	60.6
Average exits per year	86.5	114.0
Average conversions to other type per year	146.6	124.0
Bed size	98.3	109.1
Average Prop. Medicare in 2018	13.4	13.2
Average Prop. Medicaid in 2018	51.2	63.9

Note: Data in this table are sourced from our facility-level data covering 2000-2018 and includes all states.

Table 1 presents summary statistics about the facility-level data by FP status. Approximately 30% of the Medicare-certified nursing homes are not-for-profits, while this share has decreased over time. On average, facility sizes are fairly similar, with FP facilities being slightly larger. Regarding facility observables, a diverse set of organisations own NFP nursing homes. Of the 4,463 NFP providers in 2018, 953 providers are government-owned, 502 are operated by religious organisations and the remainder by private, NFP organisations. It is generally understood that for-profit nursing homes tend to operate as part of larger chains (Harrington et al., 2012), but the exact extent of this has been difficult to quantify because ownership of for-profit nursing homes is notoriously opaque.¹⁴

¹³See <https://www.councilofnonprofits.org/running-nonprofit/nonprofit-audit-guide/c/state-law-nonprofit-audit-requirements>

¹⁴To improve transparency the Biden Administration recently introduced reforms to increase reporting and monitoring of nursing home ownership (Centers for Medicare and Medicaid Services, 2023). We commenced an exercise to identify chain ownership of nursing homes in California by hand-coding. Most FP providers appeared to be part of a within-state or cross-country chain, whereas NFP providers typically appear to be independent or operate a handful of facilities at most.

Table 1 also provides summary statistics about the Medicaid and private payer shares (weighted by length of stay) at FP and NFP facilities in 2010. It's notable that across all the summary statistics, FP providers serve more Medicaid residents and less private-pay residents than their NFP counterparts. We interpret these statistics as supporting the idea that FP providers play an important role in providing access to care in needier communities. Differences in ownership and payer-types served are also suggestive of the different business models adopted by FP versus NFP nursing homes.

There is little entry and exit, with more in the for-profit sector. This may be explained by NFP providers having limited financing options: they are legally banned from accruing earnings to any private shareholder or individuals, and thus limited from equity financing. Conversions between FP and NFP nursing homes are relatively rare.

2.4 Cross-subsidisation in nursing homes

To contextualise some of our results later, it is important to understand that the US nursing homes industry is also a story of cross-subsidisation. Not only is the extent of cross-subsidisation significant, but the form also differs dramatically across NFP and FP providers. First, we consider the usage of related parties. Related parties are somewhere in between vertically integrated firms and not. An owner of a nursing home may own or have financial interests in several other businesses which are not directly vertically integrated and operate independently. For instance, in addition to the nursing home, they could have financial interests in businesses operating in catering, cleaning, medical goods, real estate and so on.¹⁵ The nursing home could contract with these related party businesses at inflated prices and through Medicaid and Medicare reimbursements which partially scale with past reported costs, extract rents from Government.

Gandhi and Olenski (2024) study this practice in the state of Illinois and find that in 2019, 68% of profits were 'hidden' using this practice, primarily through property costs and to a lesser extent management costs. While Gandhi and Olenski are the first to formally study this problem, nursing home consumer advocates and some thinktanks have thought this to be a big problem throughout the entire US nursing homes industry for some time now (The National Consumer Voice for Quality Long Term Care, 2023; Hammond, 2022). CMS cost report data allows us to have a look at the extent of usage of related parties. In 2018, about 49% and 83% of NFP providers and FP providers respectively used related party transactions. The mean and median dollar spend on related party transactions in 2018 for NFP providers is \$2.72 million and \$1.76 million, and for FP providers is \$2.79 million and \$2.03 million respectively. These figures suggest that the usage of

¹⁵See Rau (2017) (<https://kffhealthnews.org/news/care-suffers-as-more-nursing-homes-feed-money-into-corporate-webs/>) for an interesting investigative piece that provides examples of related party usage.

related parties is greater by FP providers, consistent with Gandhi and Olenski's analysis. We do not believe that this issue poses a first-order problem for our analysis. As Gandhi and Olenski (2024) show, the hidden profits arise mostly from property and management costs which can be thought of as fixed costs from the perspective of the delivery of nursing home services. In that case, pricing and quality choices are unaffected. Where there are related party transactions about nursing home services, then we may overstate costs by being unable to address this issue.

A different type of cross-subsidisation is primarily used by NFP providers and is based on other related upstream facilities which they may operate. Since nursing homes are downstream of these facilities this kind of cross-subsidisation does not pose a significant challenge to our analysis. This context is still important for understanding the business model of NFP providers, how different it is from FP providers and why NFP providers might offer a higher quality and higher priced product. These facilities are independent living facilities and assisted living facilities.¹⁶ They allow residents to live in a community with some level of supervision. Unlike nursing homes, they do not provide much in the way of medical care. Where medical care is provided, the costs may be partially or fully covered by Medicaid or Medicare. However, the majority of the cost – accommodation costs – are not. This also means that this sector is much less regulated, with little comprehensive data.

Many NFP organisations operate independent living facilities, assisted living facilities and nursing homes, often in close proximity. In these cases, the provider can offer a continuity of care contract or proposition to residents. The resident can enter the provider's care at an independent living facility or assisted living facility, and have the comfort of knowing that their own or their partner's nursing home care is sorted if they need it. Data on providers that offer continuity-of-care arrangements are relatively limited, but Zarem (2010) cites that 82% of such providers are NFP and 18% are FP. More recently, Nelson (2018) cites that 78% of such providers are NFP and 22% are FP.

The cross-subsidisation became clear when a CEO of a NFP nursing home told us in an interview that they believed that NFP providers are unlikely to acquire or construct new nursing home facilities, unless they can also acquire or construct an independent living or assisted living facility. Assisted living and independent living facilities are unregulated so we do not have any data about these, but this kind of cross-subsidisation makes sense. Independent living and assisted living facilities mostly take residents who pay the private price out-of-pocket. Also, their costs are likely to be easier to manage than for nursing homes because they don't have to provide as much medical care. Further, cross-subsidisation can also be enabled by residents who place a premium on valuing the security of being able to go to a nursing home in the same community and under the same provider as the independent living or assisted living facility.

¹⁶Independent living facilities are often referred to as retirement villages.

This does not mean that the nursing home facility is exclusive to residents of the upstream facilities however. We hand-collected data on whether each nursing home in California in 2018 was likely to be associated with an independent living or assisted living facility. We found that the average and third quartile share of Medicaid payers at nursing homes associated with such facilities is 39% and 66% respectively. By regulation a nursing home which accepts Medicaid residents at all, is not allowed to discriminate in their admission decisions.

3 Data

We use three sets of data. Most of the descriptive analysis uses data covering all US states between 2010 and 2018. Structural estimation will focus on resident stays in California from 2000 to 2010.

Resident Data The basis of our resident data is the nursing home Minimum Data Set (MDS). As per federal law (42 CFR §483.20), all nursing homes that are eligible to provide care and be reimbursed by CMS either through Medicare or Medicaid are required to complete assessments about their residents. Assessments are essentially an extensive survey that results in a dataset that provides a comprehensive ‘assessment’ of the resident’s health, mental condition and cognitive abilities. Assessments are conducted upon admission, discharge and throughout the resident’s stay at least quarterly. This data allows us to construct a panel of resident stays which is the basis of our structural estimation. It also provides zip-codes of where residents previously lived which allows us to construct zip-centroid distances.

Although the MDS also contains information about resident payer-types, it has been noted in the nursing homes literature that the payer-type fields in this data have inaccuracies (Grabowski et al., 2008; Hackmann, 2019; Cheng, 2023).¹⁷ Following the literature, we use Medicare claims data from the Medicare Provider and Analysis Review (MedPAR) files to identify when residents are covered by Medicare. To identify when residents are covered by Medicaid, we draw upon the Medicaid Analytic Extract (MAX) files. For 2001-2005 we have access to the MAX Long Term Care file and for 2006-2013, the MAX Personal Summary file. We also supplement this with data from the Medicare Beneficiary Summary file which we have from 2000-2019.

Facility Data We obtain facility characteristics and location data from LTCFocus who derive their data from the Centers for Medicare and Medicaid Services (CMS) Online Survey, Certification and

¹⁷There does not appear to be a comprehensive understanding about the source of inaccuracies, but there appear to be two types of issues. As described in Cheng (2023) (section B.2) one issue may arise from nursing home uncertainty about the payer-type. A second issue may arise from payer-types not being updated in the data from the admission assessment, even when the payer-type may have changed in a subsequent assessment (Grabowski et al., 2008).

Reporting database.¹⁸ Since data on private prices are not collected by any regulators we follow the literature and infer private prices by dividing total revenues with total quantity from California nursing homes cost report data (Huang and Hirth, 2016; Gandhi, 2023). We also source facility financial data (costs) from California cost report data.

Geographic and Demographics Data We source our demographic and geographic data from various datasets from the census bureau. It is worth noting that demographic data, including populations by age at the zipcode level only exist in the 2000 and 2010 decennial census and from the American Community Survey from 2010 onwards.¹⁹ These data exist at the county level for our whole sample period.

4 Motivating Evidence

4.1 Quality Provision

We take total nursing hours per resident-day as the quality choice of interest, following the previous literature (Hackmann, 2019; Lin, 2015, 2014; Friedrich and Hackmann, 2021). We interpret nursing hours per resident-day as a product attribute rather than an input. This reflects nursing hours being the most important determinant of resident quality outcomes and that in nursing homes, nurses play a broader role than just the delivery of medical care - they help residents with their daily lives.

Table 2 shows that FP providers are associated with lower nursing hours across all nursing types. The first row of Table 2 presents differences in unconditional means of nursing hours per resident day for each nurse type. Reading the columns from left to right, the nursing types go from higher skill, higher qualification to lower skill, lower qualification. FP providers use fewer nursing hours across both registered nurses and licensed practical nurses. They also use fewer certified nursing assistants, thus the scope for substitution away from higher skilled labour to lower skilled labour is limited. The second row presents coefficient estimates of the FP indicator when nursing intensities are regressed on a FP indicator along with facility controls and time and zip fixed effects.²⁰ We see the same pattern even allowing for time-invariant differences across zipcodes. In

¹⁸LTCFocus is a Gerontology and Healthcare Research center at Brown University. LTCFocus Public Use Data is sponsored by the National Institute on Aging (P01 AG027296) through a cooperative agreement with the Brown University School of Public Health.

¹⁹Strictly speaking, these are termed as zip-code tabulation areas by the U.S. Census Bureau, since data are not directly collected at the zipcode level.

²⁰We use zip fixed effects instead of facility fixed effects because we think it's reasonable to treat the FP status of nursing homes as exogenous and compare across FP and NFP providers in the same market. As described in section

appendix A.2, we use the same specification to explore the private prices set by nursing homes and find that NFP nursing homes in California also set higher private prices.²¹

These results are consistent with our interpretation that NFP providers offer a higher quality, higher priced product, while FP providers offer a lower quality, lower priced product. To disentangle whether these choices are driven by preferences for quality, differing costs and/or differing nonpecuniary motives we will turn to a structural model of nursing home demand and supply.

Table 2: Differences in Nursing Inputs

Hours/ resident-day	Registered Nurse	Licensed Practical Nurse	Certified Nursing Assistant
FP (Diff. in Means)	-0.322	-0.083	-0.380
FP (Regression)	-0.200 (0.009)	-0.055 (0.009)	-0.308 (0.012)
Mean of dep. var	0.492	0.853	2.26

Note: These results use yearly facility-level data across 2010-2018 and all states. Controls include facility size (in beds), aggregate assisted daily living score, which measures how dependent the population is on carer support for daily needs, aggregate case mix index, which measures how much medical care the population requires and proportion of Medicaid and Medicare residents in the facility. Year and zip fixed effects are included with standard errors clustered at the facility-level.

Throughout this paper we focus on a nursing home’s choice of nursing hours rather than quality outcomes. There are two reasons for this. First, comprehensively and accurately measuring quality outputs is difficult. Recent literature has made progress on this issue (Cheng, 2023; Einav et al., 2022); however, the quality measurement in these works are ultimately based on short-term mortality relating to a nursing home’s role as a rehabilitation center rather than as a long-term care provider. Second, the role of costs will be important in the structural model and counterfactuals. There is a clear relationship between inputs and costs, less so for outputs and costs. Still, Grabowski et al. (2013) find that NFP nursing homes provide better-quality outcomes than their FP counterparts.

2, the business models of NFP and FP providers are different and it seems unlikely that these would be changed from year-to-year to attain different benefits. This is also reflected in our data as it is uncommon for the FP status to change - see table 1).

²¹Private prices are sourced from California cost reports, thus the sample used is different to the all states data used here.

4.2 Access to Providers and Patterns of Provision

From Table 1, we already know that FP providers serve a large proportion of the market and supply many beds. Here, we focus on the fact that FP providers systematically serve different markets and different residents compared to their NFP counterparts. Table 3 column 2 presents the results of regressing the number of NFP providers minus the number of FP providers in the same zip-code regressed on zip-level demographics. NFP providers are more prevalent in richer zipcodes and less prevalent where there are larger populations of older people.

Our story for this result is that NFP providers serve a product better suited to private payers, while FP providers serve a product better suited to Medicaid payers. To build support for this interpretation we turn to columns 3-6 of table 3. Here, we compute the proportion of residents of a given payer-type amongst the population of all residents at all nursing homes of a given firm type in a zip-code and regress that on zip-level demographics. We find that both FP and NFP nursing homes serve more private-pay residents in markets with higher socioeconomic status and more Medicaid residents in markets with lower socioeconomic status.²² A 1% increase in median income is associated with approximately 11 percentage points more private payers and a decrease in Medicaid payers of a similar magnitude. These facts suggest that the spatial distribution of payer-types is correlated with socioeconomic characteristics.

To complete the story, we also need to show that FP providers prefer serving Medicaid residents and that NFP providers prefer serving private-payers.²³ In other words, that Medicaid residents are more profitable for FP providers and private-payers are more profitable for NFP providers. This requires estimation of a structural model of nursing home demand and supply, since marginal costs are not directly observed. To be clear, we do not have in mind a model where providers choose where to locate. As shown in table 1, entry is relatively rare compared to exits. Rather, the average facility is relatively old,²⁴ and location determines which facilities are more likely to stand the ‘test of time’.²⁵

²²The positive sign on the log 65 plus coefficient for private payers likely reflects a mechanical effect, that private payers are older. We observe that 80% of the private payers in our California sample are 65 or older. Younger residents at nursing homes are much more likely to be covered by Medicaid or Medicare. The former can include young people who are disabled, while the latter would include people who are receiving additional rehabilitation at a nursing home.

²³In theory, nursing homes could selectively admit residents to serve more of their preferred payer-type. However this is prohibited by regulation. Gandhi (2023) studies selective admissions by nursing homes. Our interpretation of his work is that the regulations are effective. He does find evidence that nursing homes undertake selective admissions and in particular discriminate against Medicaid-eligible residents. However, the magnitude of selective admissions which Gandhi finds is quite small. Moreover, selective admissions do not affect the arrival rate of preferred payer-types. For these reasons, we regard location patterns as more important and do not deal with selective admissions in this paper.

²⁴To the extent that most facilities were constructed many years, or even decades before our sample, we are unable to think about endogenous location decisions.

²⁵Another potential reason for the observed NFP location patterns is that NFP providers may find it easier to secure

Table 3: Location patterns

Dep. Var	#NFP - #FP	Prop. Private Payers		Prop. Mcaid Payers	
		Served by NFP	Served by FP	Served by NFP	Served by FP
log(Median Income)	0.167 (0.016)	0.108 (0.013)	0.080 (0.007)	-0.121 (0.018)	-0.141 (0.011)
log(50-64)	-0.060 (0.005)	-0.107 (0.010)	-0.032 (0.005)	0.094 (0.013)	0.052 (0.009)
log(65 plus)	-0.069 (0.005)	0.126 (0.010)	0.033 (0.005)	-0.150 (0.014)	-0.072 (0.008)
Share Black	-0.030 (0.037)	-0.046 (0.035)	-0.043 (0.011)	0.108 (0.046)	0.077 (0.018)

Note: These results use facility-level data across 2010-2018 and all states. All specifications include year and county fixed effects as well as controls for the log of the zip area and proportion of the zip which is urban. Standard errors clustered at the county level.

Whereas table 3 studies patterns relating to access from an across-market point of view, we show in table 4 that similar patterns exist within-markets. Table 4 presents the results of regressing the proportion of a given payer-type in a nursing home's population on a FP indicator, facility controls and year and zip fixed effects. Crucially, the inclusion of zip fixed effects means that we are using within-market variation to identify the FP coefficient. Even in markets where both providers of both types are operating, we see that FP providers serve larger proportions of Medicaid payers and lower proportions of Medicare and private-payers compared to their NFP counterparts. To the extent that Medicaid residents cannot afford to pay out-of-pocket, we interpret these results to suggest that even when NFP providers are located in markets with lower socioeconomic status, they still tend to serve more well-off residents. These results are again consistent with the market segmentation story where the NFP providers' product of higher quality is more suited to private payers and FP providers' product of lower quality is more suited to Medicaid payers.

donations and funding in wealthier locations. In interviews with industry members we heard that the role of donations and funding is relatively limited. In appendix A.3, we verify this claim and show that most NFP nursing homes do not receive any donations, including those in higher income markets.

Table 4: Proportion of payer-types served

	Prop. Medicaid Payers	Prop. Medicare Payers	Prop. Private Payers
FP	0.140 (0.004)	-0.060 (0.003)	-0.080 (0.003)
Mean dep. var	0.60	0.15	0.25

Note: These results use facility-level data across 2010-2018 and all states. Controls include facility size (in beds), aggregate assisted daily living score, which measures how dependent the population is on carer support for daily needs, aggregate case mix index, which measures how much medical care the population requires. All specifications include year and zip fixed effects. Standard errors clustered at the zip level.

4.3 Entries, Exits and Takeovers in California

Bans on some or all FP providers will reduce the number of nursing homes and by itself reduce welfare. Therefore, an interesting policy evaluation should allow for extensive margin changes. In response to fewer providers in the market from a ban, new NFP providers could enter. A fair evaluation of a ban should capture these effects. However, as we saw in table 1, entry activity is limited. This suggests that the fixed costs of constructing and certifying a new facility are likely to be large. Since the limited entries happen even with a larger rate of exits, this is likely to hold even in a counterfactual where some nursing homes are banned. Motivated by recent concerns about takeovers and acquisitions,²⁶ we consider the prevalence of takeovers. Lack of transparency in ownership is notorious in the nursing homes industry.²⁷ Our facility-level data assigns facility IDs based on location rather than facility-owner, thus to address this gap we undertook a hand-coding exercise to identify likely changes in ownership. This was done on the basis of changes in facility name, administrator, profit-status and googling facility histories. We summarise the results below:

²⁶Gupta et al. (2021) find that the wave of private equity takeovers increased mortality in nursing homes. Harrington et al. (2017) notes that these patterns of FP takeovers are common in many countries and are often associated with quality deficiencies.

²⁷This has led to recent improvements in regulation of ownership, with the Centers for Medicare and Medicaid services requiring more data and disclosures about nursing home ownership (Centers for Medicare and Medicaid Services, 2023)

Table 5: Entries, exits and takeovers in California, 2000-2018

	New facility	Facility closure	Own type takeover
FP	36	130	523
NFP	13	117	4

Note: There are very few across firm type takeovers. 21 NFP becoming FP and 2 FP to NFP.

The relative magnitude of takeover versus entry activity amongst FP providers is striking. In appendix A.5, we check that our hand-coded takeovers do reflect true takeovers by looking for evidence of changes in management. We find that there is a significant shift in the payer-type population of FP facilities which undergo takeovers, suggestive of a change from management. Gupta et al. (2021) explicitly study takeovers by private equity. While they do not look at changes in payer-type populations, we do find similar results on nursing hours and costs. To be clear, we do not regard the effect of the takeover as important (except that there is one) as our counterfactual focuses on FP facilities being taken over by NFP providers. In-sample takeovers are of little relevance except that they highlight that the relevant margin of extensive margin activity for the counterfactual should be from takeovers rather than the construction of new facilities.

5 Structural Model

There are two reasons why we develop and estimate a structural model. First, we are interested in studying counterfactual outcomes when we vary the number of NFP and FP providers. To the best of our knowledge, there are no satisfactory policy changes with exogenous variation in the number of NFP versus FP providers which would allow us to explore such counterfactuals. Second, we aim to explain the patterns of quality, access and provision across FP and NFP established in the motivating evidence. There are a range of different features of the market that could potentially lead to observed patterns including heterogeneous preferences for quality and private prices, provider non-pecuniary motives and provider costs. A sufficiently flexible structural model is required to coherently distinguish between these potential causes.

In addition, regarding access, we showed that NFP providers are more prevalent in markets with higher socioeconomic status and FP providers are more prevalent in markets with lower socioeconomic status. We also showed that providers serve more Medicaid and less private payers in markets with lower socioeconomic status and vice-versa for markets with higher socioeconomic status. To complete the argument that the location patterns are explained by NFP and FP providers preferring particular payer-types, we need to examine the margins that providers make on each

payer-type. We estimate these with a structural model.

The core of our model is a static oligopoly game where providers compete by choosing their private price p_{jt} and quality (total nursing hours per resident-day) TN_{jt} ²⁸. Markets will be defined by metropolitan statistical area-year pairs in California from 2000-2010. Providers draw their cost and demand shocks before setting prices and quality, and these are all observed by providers, thus the equilibrium concept is a Nash equilibrium. Following this, residents choose the nursing home which maximises their utility.

Before spelling out the details, we provide a quick preview to highlight the important features. Providers have standard motivations from profit, but they may also have nonpecuniary motives. Specifically, we allow for providers to explicitly place value on the quantity-of-care they serve, and this will be a parameter to be estimated. If residents value quality for example, then a nonpecuniary motive for quantity-of-care served could lead to the provider choosing higher quality as well. This necessitates the estimation of demand. Our specification of demand is rich allowing for substantive heterogeneity in preferences for private prices and quality. This could generate incentives for providers to segment and target particular payer-types in their pricing and quality choices. Provider costs are also relevant. Different payer-types may have different costs (and margins) and these costs may also differ by firm-type. Whether a firm would prefer to choose a higher quality depends on in addition to their objective, whether higher quality attracts more desirable, higher-margin residents. This justifies the estimation of a rich supply model and can also explain the market segmentation we document. Modelling payer-type margins not only allows us to explain quality choices, but also the patterns of access we document. In addition to choosing prices and quality to target particular payer-types, firms in markets with more of the desirable payer-types will be relatively profitable and more likely to operate there.

6 Demand

We adopt a similar model of demand as Hackmann (2019). Prior to a potential stay, resident i chooses the nursing home j that maximises their daily utility during the stay.²⁹ Individual nurs-

²⁸We model providers who choose their quality by choosing their total nursing hours per resident-day. This follows previous structural work such as Hackmann (2019) and Lin (2015) and a broader literature on the effects of nursing inputs on quality outcomes (Lin, 2014; Friedrich and Hackmann, 2021). We also follow the literature and restrict nursing hours to focus exclusively on higher-skilled nurses - registered nurses and licensed practical nurses but not certified nursing assistants.

²⁹Residents implicitly choose the nursing home which maximises the utility of their entire stay which is the product of their daily utility and length of stay. Implicit in this, we assume that length of stay is exogenous. On the demand side this means that residents do not evaluate how their length of stay might change if they chose a different nursing home. On the supply side, this means providers' choices cannot affect the length of stay through their quality choices.

ing home stays need not adhere to a calendar year and 22% of stays in our data overlap with at least 1 year. We assume that the nursing home commits to the same price and quality throughout a resident's stay. That is, in subsequent years, the nursing home's pricing and quality choices apply only to residents entering in that year. Since residents only decide at the start of their stay, allowing for prices and quality to change for the resident in subsequent years would result in having to treat the resident as fully 'locked in' for subsequent years.³⁰ Residents who are only Medicare or Medicaid payers over the course of their stay never pay the private price, while those who are private-payers at some stage do. Accordingly, we further distinguish resident utility by $y \in \{\text{non-priv}, \text{pays priv}\}$.

³¹ For brevity we will call y a payer-type as well when there should be no ambiguity. Then an individual's utility is:

$$u_{ijt}^y = \beta_{1i}^d D_{ij} + \beta_{2i}^d D_{ij}^2 + \beta^I \mathbf{1}(j = j(t-1)) + \beta_i^{TN} TN_{jt} + \beta_i^{p,y} p_{jt}^{priv} + \beta_i^{NFP} NFP_j + \lambda_j + \lambda_t + \xi_{jt}^y + \varepsilon_{ijt}$$

Coefficients which vary by individual are of the form:

$$\beta_i^k = \beta^k + \sum_r z_{ir}^k \beta_r^k$$

With $\beta_i^{p,\text{non-priv}} = 0$.

Residents have preferences for distance D_{ij} which are allowed to be nonlinear. We allow preferences for distance to vary by age of the resident on entry, realised length of stay and whether the resident has a legal guardian who is not themselves - this could be a family member or a lawyer.³² We allow for inertia in demand through $\mathbf{1}(j = j(t-1))$, reflecting high switching costs once a resident has moved into a nursing home. Since our data span a long period (2000-2010), many residents undergo multiple stays. Inertia could introduce dynamic considerations however, we assume that residents are myopic so that they do not take into account how inertia could affect their future decisions. The presence of inertia could introduce invest-harvest incentives for nurs-

³⁰Since private price setting is largely unregulated, and nursing homes do not typically advertise their private price rates, to the best of our knowledge it is not clear what happens in practice.

³¹Although Private-pay, Medicare and Medicaid make up the vast majority of residents, some residents are of a different payer-type altogether. Two examples of such payer-types are those who access nursing home care through Veteran Affairs or a Medicare/Medicaid managed care program. The facility data reports that the median and average proportion of resident-days from such payer-types are 5.7% and 9.2% respectively. In both these cases, residents face restrictions on their choice set which we do not observe. We also don't know the prices that nursing homes get from these residents. For these reasons, we do not model demand from these residents and treat them as exogenous on the supply side.

³²For length of stay, it would be ideal to use an anticipated length of stay instead of actual length of stay to correspond more closely with a resident's decision making. While the MDS2.0 contains an anticipated stay variable, we find that there are too many missing cases for this to be usable. We incorporate realised length of stay by bucketing stays into 3 groups. Stays within 30 days, stays from 30-100 days and stays of 100 days or over. The 30 day choice coincides with bucketing used for the anticipated length of stay in the MDS data, while the 100 day choice corresponds to the Medicare limit.

ing homes to price and choose quality dynamically. We are not aware of any papers or reports with these concerns, thus we abstract from these possibilities in modelling.

Preferences for quality (as measured by total nursing hours per resident day) vary by whether the resident is a private, Medicare or Medicaid payer, length of stay and a large range of other demographics.³³ Preferences for price vary with whether a resident pays the private price on entry, their age, their length of stay, whether they have at least a bachelor's degree and whether they have a legal guardian who is not themselves. Allowing for significant heterogeneity in preferences for quality and private price supports the ability of the model to explain market segmentation through demand heterogeneity. Nursing homes may choose different private prices and qualities to avoid competing across similar pools of residents.

We also allow residents to have preferences for NFP providers. This preference captures the idea of NFPs as solving information asymmetry problems. If quality is unobserved or hard to verify, residents may prefer NFP providers as their incentives may be more closely aligned with their own residents' (Weisbrod, 1988). Although residents in our model observe a nursing home's total nursing hours per resident day, there may be other aspects of quality that a resident has imperfect information about. For example, this could include how well-trained nurses are for the nursing home. Differences in quality that are unobserved to the econometrician and systematic across NFP versus FP may also be captured in this preference. For example, our data on nursing homes does not include information about the quality of the accommodation. If NFP provide better-quality accommodation on average than FP providers, and residents value this, it would be picked up in this preference for NFP providers. Given this interpretation of NFP preferences, we allow these preferences to vary with the same set of individual characteristics as with observed quality (total nursing hours per resident day).

λ_j are zip-fixed effects and λ_t are year fixed effects. For the estimation of mean preference parameters, we use instruments that are weighted by distances between nursing homes. These distances will be larger on average in less dense areas which could correlate with demand. Conditioning on zip fixed-effects mitigates this concern. ξ_{jt} are the unobserved demand shocks and ε_{ijt} are the standard type 1 EV idiosyncratic demand shocks. For estimation we can define nursing home mean utilities:

$$\begin{aligned}\delta_{jt}^{\text{priv}} &= \alpha^{TN} TN_{jt} + \alpha^{p.\text{priv}} p_{jt}^{\text{priv}} + \alpha^{NFP} NFP_{jt} + \lambda_j + \lambda_t + \xi_{jt}^{\text{priv}} \\ \delta_{jt}^{\text{non-priv}} &= \alpha^{TN} TN_{jt} + \alpha^{NFP} NFP_{jt} + \lambda_j + \lambda_t + \xi_{jt}^{\text{non-priv}}\end{aligned}$$

Although markets are defined by metropolitan statistical areas, for tractability, we restrict a resi-

³³This includes age, a resident's depression rating scale score - a measure of mental health, cognitive performance scale score, assisted daily living score, clinically complex scale score, whether they have alzheimer's, are female, are nonwhite, have at least a bachelor's education and whether they have a legal guardian who is not themselves.

dent's choice set to all nursing homes within 30 miles of their prior residence. As we saw in figure 1 this allows us to capture 95% of residents, with the remaining tail omitted from estimation. We also make three types of exclusions for facility-years, where excluded facility-years are pooled into an outside option with utility normalised to 0.³⁴ First, we assume that any data are missing at random and exclude facility-years if there is a missing observable. Second, we drop a facility-year if there are less than 5 never-private payers or 5 residents who pay the private price during their stay. Finally, we drop facility-years where the proportion of Medicare bed-days served exceeds 90% following Hackmann (2019); Feng et al. (2008) to avoid modeling resident choices where extensive and expensive rehabilitation is required at highly specialized facilities. After these exclusions, our demand estimation sample is comprised of around 940,000 nursing home stays.

Estimation and Identification

We estimate demand using a two-step procedure following Hackmann (2019). In the first step, we recover $(\delta_{jt}^y, \beta_i^k, \beta_{1i}^d, \beta_{2i}^d, \beta^I)$ by Maximum likelihood. As is standard in demand estimation, conditional on δ_{jt}^y , the remaining variation in price and quality is exogenous. In the second step, we recover the remaining mean preference parameters with a generalised method of moments estimator. Private prices and quality are correlated with ξ_{jt}^y thus we need to instrument for them.

We include in our instruments four types of excluded cost shifters. First, we use the distance-weighted number of FP and NFP providers in the market. The number of FP and NFP providers in the market is an instrument based on excluded product characteristics. Conditional on zip and year fixed-effects they shift the nursing home's price and quality, but are not correlated with ξ_{jt}^y . Second we use the distance-weighted average proportion of Medicaid and Medicare residents across other facilities. Different groups of residents have different care needs and thus, different costs. The proportion of Medicaid and Medicare residents are at the market-level, and distance between nursing home facilities is exogenous, thus conditional on zip and year fixed-effects, these instruments should not be correlated with ξ_{jt}^y . Third we use the nursing home's Medicaid per resident-day reimbursement rate. These rates are set prospectively by CMS, thus conditional on zip and year fixed-effects they should not be correlated with demand shocks. On the other hand, they shift a nursing home's quality choice as they affect the profitability of Medicaid residents (Hackmann, 2019). Finally, we use the distance-weighted sum of registered nurse hours and distance-weighted sum of nursing assistant hours across other facilities. These measures capture the local availability of nurses which affects the nursing home's costs, but not the nursing home's own demand, conditional on zip and year fixed effects.

³⁴Note that the universe of residents are those who choose a nursing home (but not necessarily one we observe). That is, the outside option does not include choices made by old people who choose not to go to a nursing home.

Results

Table 6 presents key parameters in our demand estimates. A full list of parameter estimates with demographic interactions can be found in table 16 in the appendix. We summarise the variation in estimated distance, price and quality elasticities in figure 2. We also include the distribution of estimated preferences for NFP providers (β_i^{NFP}).

Table 6: Main demand estimates

Coefficient	Estimate
Distance (miles)	-0.1195 (0.0009)
Distance squared	-0.0001 (0.00007)
Inertia	3.9590 (0.0037)
Private Price	-0.0248 (0.0051)
Total Nursing	-1.3257 (0.7248)
Total Nursing x Medicaid Payer	-0.3370 (0.0063)
Total Nursing x Medicare Payer	0.6284 (0.0041)
Total Nursing x Age	0.0235 (0.0003)
NFP	-4.0118 (0.2112)
NFP x Medicaid Payer	-0.5269 (0.0109)
NFP x Medicare Payer	0.8267 (0.0049)
NFP x Age	0.0514 (0.0008)

Note: These results use individual-level data from 2000-2010 in California. First stage parameters are estimated by maximum likelihood using a choice panel with around 77 million nursing home - individual observations. Second stage parameters (baseline preference for private price, total nursing and NFP) are estimated by TSLS with year and zip fixed effects with 12,990 pays private-price/never pays private price - nursing-home - year mean utilities. F-statistics for the first stage of the TSLS are 116.06 for total nursing hours per resident day and 20.10 for private price. Standard errors computed by block (drawing facilities) bootstrap with 400 iterations.

Broadly, we obtain sensible signs and magnitudes for our demand estimates. Our distance parameter estimates suggest that residents dislike being at a nursing home far away from their prior residence. The average distance elasticity is -1.31 which is considerably smaller than those

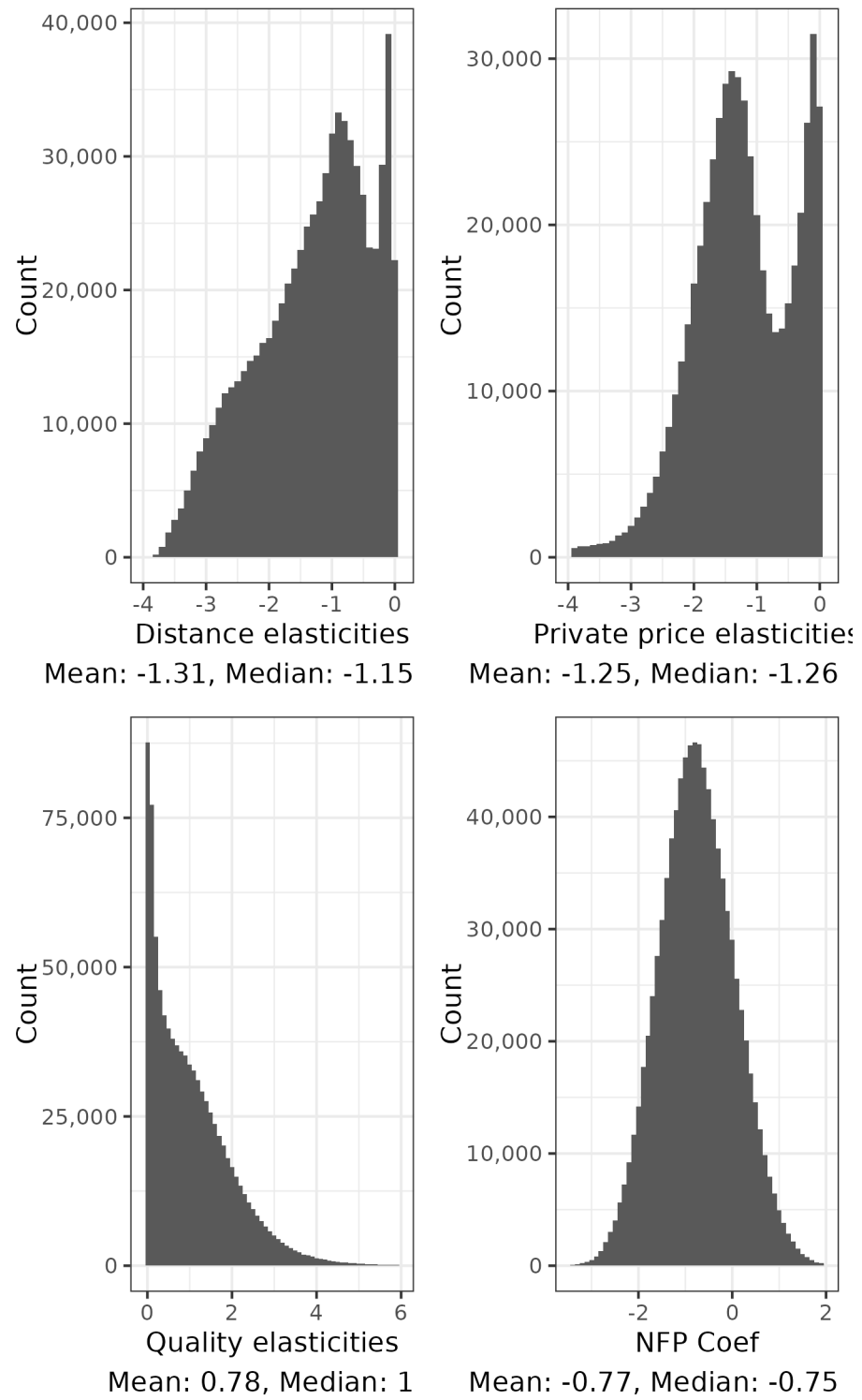


Figure 2: Individual demand elasticities

found in Gandhi (2023), who also studies the Californian market like we do. This is likely because we truncate resident choice sets at 30 miles rather than 20 kilometers, thus our data contains a stronger tail of residents who travel further than the data used in Gandhi (2023). Still, figure 2 also shows many residents who are more sensitive to distance than the average resident.

The inertia parameter estimates indicate that residents have a strong preference for staying at the same nursing home if they have multiple nursing home stays. This is unsurprising, residents who stay in nursing homes for long periods develop relationships with staff and other residents and a change in a nursing home would disrupt their medical care as well. It is worth highlighting that the relative size of this parameter, means that inertia also plays a substantive role in the counterfactuals. In particular, this implies that following a takeover, the provider has some latitude in adjusting price and quality, as inertia dampens the need to compete.

Residents dislike the private price if they have to pay. The average price elasticity is -1.25. In this case, the average is somewhat misleading as the histogram shows significant masses of residents with elasticities on either side of -1.25.³⁵ As shown in the appendix, we find that residents who enter the nursing home as a private payer are more price sensitive, consistent with the immediate saliency of private prices. We also find that residents who are at least bachelor educated are less price sensitive.

Residents value quality (total nursing hours per resident day). The histogram shows that preferences for quality are almost entirely positive.³⁶ The negative baseline parameter estimate likely arises because of the quality and age interaction.³⁷ On average, we obtain a quality elasticity of 0.78% which is comparable to those in Gandhi (2023). Again there is significant variation in the quality elasticities. Two systematic patterns we highlight here are that those who are a Medicaid payer at the time of entering a nursing home have a lower preference for quality compared to their private payer, and especially, Medicare counterparts. This is consistent with the observed market segmentation, where NFP providers serve more private payers and higher quality, while FP providers serve more Medicaid payers and lower quality. In the appendix, we find that residents with higher medical and nursing needs such as those with higher ADLs and CLNs, and those who are at least bachelor educated have stronger quality elasticities, on the other hand those with weaker decision making capabilities such as those who are more depressed, cognitively challenged, with Alzheimer's or who have someone else making decisions have weaker preferences for quality.

³⁵The price elasticity histogram is truncated on the left at -4. Around 0.6% of stays have a price elasticity of more than 4.

³⁶The quality elasticity histogram is truncated at the left at 0. Around 5% of resident-stays have a negative quality elasticity

³⁷We include age as a continuous variable to reduce parameters. Since we restrict our sample to residents aged 65 and above, it is more useful to think of the baseline parameter estimate as $-1.3257 + 65 \times 0.0235 = 0.2018$

On average, there is a bias against NFP nursing homes, although there is still a considerable tail of residents who prefer NFP providers. This could reflect two things. First, conditional on nursing homes with the same total nursing hours, residents are not willing to pay for any additional amenities which a NFP provider might provide. Second, we do not model capacity constraints and provider selection of residents which can be important on the margin as in Gandhi (2023).³⁸ NFP providers are smaller on average compared to FP providers, thus residents may be less likely admitted into a NFP nursing home on average. Overall, we think it's reasonable to interpret much of these estimates as reflecting preferences. Residents with the longest stays are more likely to prefer NFP providers, which as Gandhi finds are the most likely to be rejected for admission. We also find other systematic shifters of preferences for NFP providers that are reasonably similar to systematic shifters of preferences for quality such as those with at least a bachelor's education being more likely to prefer NFP providers. A reasonable interpretation of these results is that residents who have a more 'salient' nursing home stay will care more about dimensions of quality that are harder to observe such as accommodation and food quality and therefore value NFP providers who on average do better on these dimensions of quality. This interpretation concords with our suggestive evidence discussed with our supply results, that NFP providers spend more on these other dimensions of quality.

7 Supply

In each market, providers first observe their unobserved demand and cost shocks and then compete by choosing their private (per-day) price and total nursing hours in a static oligopoly. To setup the nursing home's problem we first need to aggregate individual choice probabilities using our estimated demand system into firm quantities. Ignoring the m subscript for brevity, let $q_{jt}^\tau := \sum_i s_{ijt}^\tau \text{Days}_i^\tau$ be the quantity of resident-days for each payer-type served, $Q_{jt} = \sum_\tau q_{jt}^\tau$ be the total quantity of resident-days served across all payer-types and $f \in \{FP, NFP\}$. Note that for the empirical model, we effectively use facility-reported payer-type quantities, which do not necessarily match the quantities aggregated from our demand data. This can arise for various reasons such as residents with missing observables. The nursing home's problem is:

$$\max_{TN_{jt}, p_{jt}^{priv}} \underbrace{\sum_{\tau=\text{mcaid, mcare, priv, ins}} q_{jt}^\tau p_{jt}^\tau}_{\text{Total Revenue}} - \underbrace{C^f(q_{jt}^\tau, TN_{jt})}_{\text{Total Costs}} + \underbrace{\alpha_Q^f Q_{jt}}_{\text{nonpecuniary motives}}$$

³⁸Gandhi (2023) estimates a structural model of admissions to California skilled nursing facilities to study selective admission practices. He finds evidence that facilities disproportionately exclude Medicaid-eligible patients with long anticipated stays. Still, he finds that even nursing homes close to full occupancy, admit Medicaid-eligible patients with around 70% probability. Thus we think it's reasonable to abstract from selective admissions and focus on pricing and quality choices instead for this paper.

Providers choose their total nursing hours and private prices to maximise an objective which depends on both the profits they make and the quantity-of-care they provide. Total revenues are the aggregate of revenues from each payer-type. ‘priv’ and ‘ins’ refer separately to payer-days where the resident pays the private price and ‘ins’ when the resident’s stay is covered by long-term insurance. As long-term insurance makes up a small proportion of stays - 6% - we do not model these distinctly and will think of such stays as being funded by a program like Medicare or Medicaid where the resident does not pay the private price.³⁹

α_Q^f governs how much providers of a particular organisation type value providing quantity in equivalent profits and are parameters to be identified and estimated. They vary by f meaning that a separate parameter for each organisation type will be estimated. These parameters capture two non-pecuniary motives. Providers may care about delivering care to more residents, but since residents have preferences for quality, it can also capture motives for delivering higher quality care. An alternative approach would be to follow Newhouse (1970) who describes a model of NFP providers who have both quality and quantity-of-care non-pecuniary motives. Since there are already many supply parameters to be estimated and separately identifying two distinct conduct parameters is especially challenging, we opt for parsimony and focus on a quantity-of-care motive only. Other papers often capture other motives by assuming that firms care about consumer surplus in addition to profits. We decided against using consumer surplus for two reasons. First, using consumer surplus also implies that firms consider willingness to pay in their nonpecuniary motive. We think it is more plausible that nursing homes, particularly NFP nursing homes, would simply weight residents equally in their nonpecuniary motive. Secondly, the interpretation of consumer surplus for both the econometrician and nursing home is not as straightforward as not all residents face the private price.

Firm cost functions are as follows:

$$C^f(q_{jt}^\tau, TN_{jt}) = \underbrace{Q_{jt}(\gamma_0^f TN_{jt} + \gamma_2^f K_{jt} + \gamma_3^f K_{jt}^2 + \mu_j + \mu_t + \eta_{jt})}_{\text{Costs common across payer-types}} + \underbrace{\sum_{\tau=\text{mcaid, mcare}} \phi_0^{\tau, f} q_{jt}^\tau + \phi_1^{\tau, f} q_{jt}^\tau TN_{jt}}_{\text{Costs arising from payer-types}} + \underbrace{U_{jt} TN_{jt}}_{\text{Unobserved quality fixed costs}}$$

Cost functions differ systematically across firm types which allows us to explore differences in firm efficiencies. Nursing homes incur common costs across payer-types, including a common total nursing hours cost, and the variable costs of operating the facility captured by K_{jt} the size of the facility (allowing for economies of scale) and zip and year fixed effects μ_j, μ_t . We use fixed

³⁹ Analogously, in demand estimation we treat a resident who is covered by long-term insurance as not paying the private price.

effects at the zip level instead of the facility level because we have relatively few observations per facility on the supply side. Finally η_{jt} captures unobserved facility marginal cost shocks.

The second component of costs allows nursing home costs to vary with the distribution of residents across payer-types. As nursing homes serve more Medicaid and Medicare patients they may incur higher or lower marginal costs. $\phi_1^{\tau,f}$ also allows nursing costs to vary with Medicaid and Medicare patients. Total nursing hours per resident day captures all types of licensed nurses including higher skilled and higher cost registered nurses. Nursing homes which serve more Medicare patients for example could require more registered nurses compared to licensed practical nurses due to the higher medical care needs. This would lead to a steeper quality cost gradient. This specification of costs allows us to tractably capture payer-type margins that vary systematically across firm type. They are important for explaining the patterns of access we documented previously. The concept also accords with our interviews with industry where it was common to see them discuss the ‘margin’ on each payer-type. Finally the last term captures unobserved costs only associated with quality. The latter is needed to rationalise a firm’s quality choices with the data through the quality first-order condition, as in Fan (2013).

Estimation and Identification

Having described the supply model, we can discuss identification. It is useful to consider the first-order conditions abstracting away from the cost function:

$$\begin{aligned} [p_{jt}^{priv}] : \frac{\partial}{\partial p_{jt}^{priv}} \left(\sum_{\tau} q_{jt}^{\tau} p_{jt}^{\tau} \right) &= -\alpha_Q^f \frac{\partial Q_{jt}}{\partial p_{jt}^{priv}} + \frac{\partial C^f(q_{jt}^{\tau}, \eta_{jt})}{\partial p_{jt}^{priv}} \\ [TN_{jt}] : \frac{\partial}{\partial TN_{jt}} \left(\sum_{\tau} q_{jt}^{\tau} p_{jt}^{\tau} \right) &= -\alpha_Q^f \frac{\partial Q_{jt}}{\partial TN_{jt}} + \frac{\partial C^f(q_{jt}^{\tau}, \eta_{jt}, U_{jt})}{\partial TN_{jt}} \end{aligned}$$

The left-hand side of these first-order conditions are the marginal revenues from adjusting price or quality and are known given our demand system.⁴⁰ As usual, we can invert these equations, project them onto the observables and recover parameters with assumptions on the unobserved cost shocks η_{jt}, U_{jt} .

This representation also highlights that α_Q^f can be viewed as rotating marginal revenues outwards. When $\alpha_Q^f > 0$, firms have nonpecuniary incentives for serving a higher quantity and are willing to incur a higher level of marginal cost. Therefore, the identification challenge with α_Q^f is to distinguish between higher levels of marginal cost arising from firms with more inefficient

⁴⁰Note that in general, $\frac{\partial q_{jt}^{\tau}}{\partial p_{jt}^{priv}}, \tau = \{mcaid, mcare\}$ need not be 0 due to residents who transition to and from paying the private price.

marginal cost function parameters, or firms which care more about providing quantity-of-care.⁴¹ We follow a similar idea as in Hackmann (2019) and Kim (2024) and match the mean of simulated variable costs with the mean of actual variable costs:⁴²

$$\mathbb{E} \left[VC_{jt} - \hat{VC}_{jt}(\theta_s^f) \right] = 0,$$

Where VC_{jt}^f are variable costs from cost report data, and $\hat{VC}_{jt}^f$ are simulated variable costs. We match variable costs because unlike those two papers, our cost function exhibits nonlinearities from $\phi_0^{\tau,f}$ and $\phi_1^{\tau,f}$. Thus, we cannot compute marginal costs from data (as say, average variable cost) and directly match marginal costs. Since α_Q^f does not affect variable costs it rationalises the difference between simulated and observed variable costs.⁴³ Finally, in constructing this moment we winsorize the data at the 98th percentile for average variable costs. This avoids matching on especially high variable cost reports. Following this idea of matching costs, we add two additional moments. First, we match on the within-facility first-difference in variable costs:

$$\mathbb{E} \left[(VC_{j,t} - VC_{j,t-1}) - (\hat{VC}_{j,t}(\theta_s^f) - \hat{VC}_{j,t-1}(\theta_s^f)) \right] = 0,$$

While our focus on variable costs should allow us to avoid the bulk of the tunnelling concern raised in Gandhi and Olenski (2024), our construction of variable costs is based on cost center definitions and therefore quite broad. First-differencing variable costs removes level (fixed) components. In other words, variation in this moment is driven entirely by changes in variable costs arising from changes in the quantity of patient-days served. Second, we can also think of α_Q^f as analogous to a conduct parameter, in the spirit of the classic conduct-testing literature (e.g., Bresnahan, 1982; Berry and Haile, 2014). The intuition is that excluded demand rotators shift marginal revenue while leaving costs unchanged. Since different values of α_Q^f imply different mappings from marginal revenue to recovered marginal costs, these rotators generate variation that helps distinguish objective-function parameters from cost-function parameters. We therefore impose the additional moment conditions:

$$\mathbb{E} \left[Z_{jt}^{VC} \left(VC_{jt} - \hat{VC}_{jt}(\theta_s^f) \right) \right] = 0,$$

$$\mathbb{E} \left[Z_{jt}^{VC} \left((VC_{j,t} - VC_{j,t-1}) - (\hat{VC}_{j,t}(\theta_s^f) - \hat{VC}_{j,t-1}(\theta_s^f)) \right) \right] = 0,$$

⁴¹Specifically, since α_Q^f is a level shifter of marginal costs we cannot distinguish it from μ_j, μ_t without an additional source of variation.

⁴²Constructing variable costs from the data requires taking a stance on what is 'variable'. We use the categories of operating expenses by cost center in the cost report data adding the following categories: Skilled Nursing Care, all Ancillary services except Home Health Services and all support services except Administration. We do not include property expenses, interest expenses and provisions for bad debts which are likely to be fixed costs and subject to the tunnelling concerns discussed in Gandhi and Olenski (2024).

⁴³Since quality fixed costs are fixed from the perspective of quantity they are also not included in the matching of variable costs.

where Z_{jt}^{VC} is the excluded demand rotator. The specific rotator we use is the number of Medicare-eligible individuals at the zip-code level, which affects market size and demand for nursing home care, but should not be correlated with unobserved cost shocks.

The remaining identification challenges are standard. Since the firm chooses TN_{jt} and p_{jt}^{priv} knowing η_{jt} and U_{jt} , they are likely correlated. Thus we need instruments Z_{jt} which shift TN_{jt} and p_{jt}^{priv} but are not correlated with cost otherwise.⁴⁴ With instruments Z_{jt} in hand, our moment conditions are:

$$\mathbb{E}[Z_{jt}\eta_{jt}] = 0, \mathbb{E}[Z_{jt}U_{jt}] = 0$$

We estimate supply side parameters by GMM using the moments above, separately for FP versus NFP providers.⁴⁵ This means we do not impose any further parametric structure on how costs between NFP and FP providers can differ.

Our rich cost function specification has many endogenous variables for each moment condition. Accordingly, many instruments are required. The first set of instruments are those relating to competitors. The number of distance-weighted FP and NFP nursing homes, the distance-weighted sum of total beds and distance to nearest nursing home. All of these relate to exogenous characteristics about competitors which are not correlated with the nursing home's own η_{jt}, U_{jt} . These should shift the quantities that the nursing home faces and the quality they choose in response to those exogenous factors.

The second set of instruments relates to discharges to nursing homes from other medical facilities. Discharges to nursing homes from other types of medical facilities are unlikely to be correlated with the nursing home's own η_{jt}, U_{jt} , but shift the quantities they are likely to serve. We use the distance of a nursing home to the nearest hospital, as hospitals can discharge residents directly to nursing homes.

We also look at home-health discharges. Home-health services provide nursing care to residents in their own home instead of at a nursing facility. Since nurses cannot typically be at someone's house 24/7, people who find themselves needing more intensive supervision can be discharged to a nursing facility. We have home health data that provides the total number of discharges to intermediate care facilities or skilled nursing facilities at a facility-year level. This enables us to construct distance-weighted discharges for a particular nursing home. The data we use also breaks down home health quantities by payer-type (private-payer, Medicare, Medicaid), but not specifically the number of discharges to particular nursing homes. Since this directly speaks

⁴⁴We treat K_{jt} as exogenous. In the data we observe very few changes in capacity over time. Across all providers and years we only see expansions of capacity in 3.4% of cases and contractions of capacity in 4.4% of cases.

⁴⁵In constructing the sample moment conditions, it is worth noting that U_{jt} does not show up in the pricing first order condition. Thus for a given guess of parameters, we can compute η_{jt} and then use the quality first order condition to back out the implied U_{jt}

to our payer-type quantities we also include these distance weighted home-health quantities. All else equal a higher quantity served at a home health facility also results in more discharges.

Results

Table 7: Supply Parameters

	NFP	FP
Quantity Motive	28.34 (13.66)	-9.55 (23.89)
Quality Cost (Baseline)	106.22 (146.87)	42.27 (20.02)
Quality $\times$ Mcaid	-131.55 (192.42)	-52.01 (25.05)
Quality $\times$ Mcare	0.91 (330.11)	77.78 (107.22)
Mcaid Quantity	285.22 (300.59)	136.38 (67.13)
Mcare Quantity	137.95 (690.06)	4.27 (536.93)
Facility Size in 100 beds	-50.18 (83.82)	-58.94 (23.97)
Facility Size (Quadratic) in 100 beds	19.73 (45.24)	15.59 (6.87)
N	747	5615

Note: Parameters can be interpreted in \$/bed-day. Zip and year fixed effects included. These results use facility-level data from 2000-2010 in California. Standard errors computed using the delta method, combining the supply-side GMM (clustered at the firm level) and the block bootstrap using 400 iterations from the demand-side.

Table 8: Supply Instrument F-Statistics

	Quality	Total Quantity	Medicaid Quantity	Medicare Quantity
FP	16.42	71.30	61.17	23.29
NFP	6.67	73.89	91.70	25.01

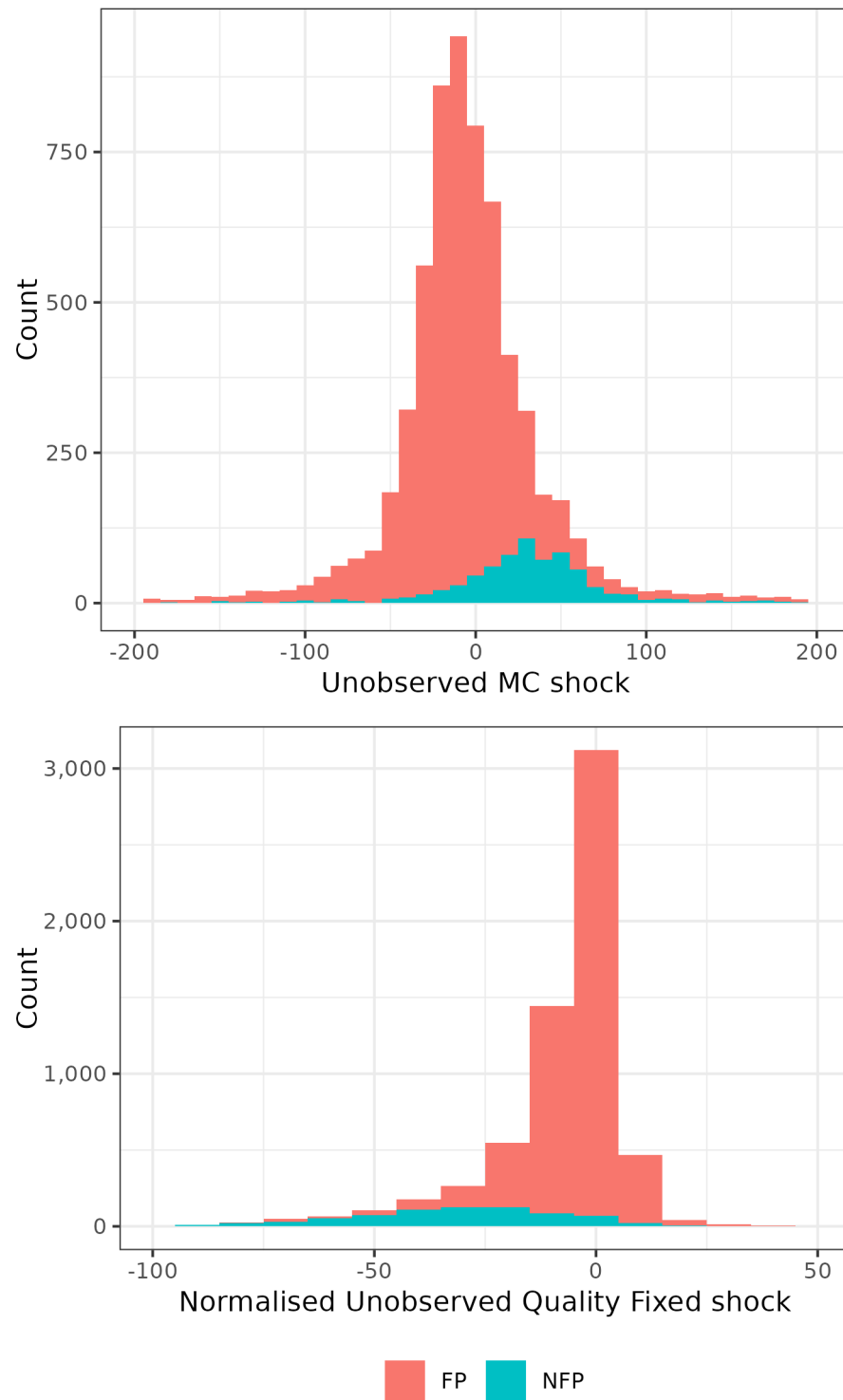


Figure 3: Realised unobserved cost shocks

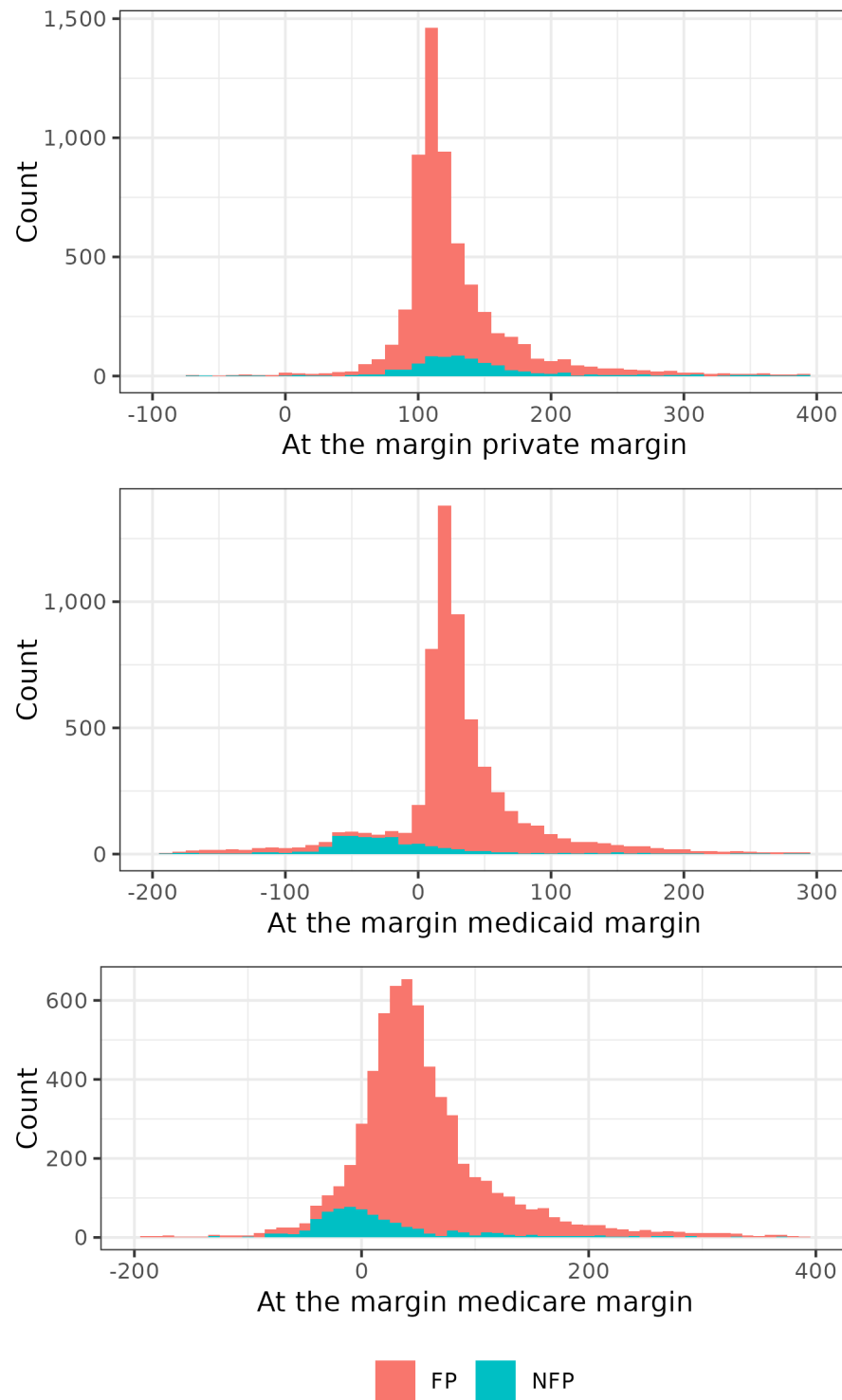


Figure 4: Payer-type margins, at observed quantities

We begin with our supply parameter estimates in table 7. All of these can be interpreted in terms of dollars per bed-day. Table 8 presents F-statistics from analogous first-stage regressions of the endogenous variables across FP and NFP providers. The objective parameter estimates indicate that NFP and FP providers value providing an additional bed-day of care at approximately \$28 and -\$10 per bed-day respectively. These results are consistent with regulations requiring NFP providers to provide public benefits in return for their generous tax treatment. The result for NFP providers appears to be of a similar magnitude as in Hackmann (2019). Hackmann (2019) considers government and private NFP nursing homes separately and estimates that they act as if they have a marginal cost advantage of \$36 and \$23 respectively. To gauge the relevance of the \$28 estimate, note that the average NFP private price is approximately \$188 per bed-day so this represents about 15% of the price or one-third of their average margin which is \$84 per bed-day.

Turning to quality costs, our estimates suggest that NFP providers have larger baseline (private payer) quality costs than FP providers. Our cost parameters for NFP providers are quite noisy compared to FP providers reflecting the relative sample sizes and the relative weakness of the instruments for quality. Still we expect to see NFP providers having larger quality costs compared to FP providers as we saw that NFP providers use a significantly higher skilled mix of nursing hours compared to FP providers (table 2). Quality costs are declining with Medicaid payers and increasing with Medicare payers for both NFP and FP providers. This is consistent with a lower skilled mix of nursing hours being correlated with more Medicaid payers and less Medicare payers being served as documented in the appendix.

To best understand how costs vary across types of providers and payer-types, table 9 calculates model implied marginal costs. Across payer-types our estimates suggest that Medicaid and Medicare payers have similar costs. While Medicare payers may require higher care needs than their Medicaid counterparts leading to higher nursing costs, our estimates suggest that costs not captured by direct nursing hours for Medicaid payers roughly equalise the overall cost. Since Medicaid payers have significantly longer stays than Medicare payers on average, nursing homes which serve more Medicaid payers need to invest and spend more on non-medical, accommodation and quality of care spending. Our estimates also suggest that private payers are significantly less costly than Medicaid and Medicare payers to serve. It is plausible that private payer days are less costly for nursing homes to serve. Comparing the average resident who has a larger proportion of their stay as a private payer than as a Medicaid payer to those with a smaller proportion, they are less than half as likely to have Alzheimer's, have a half a standard deviation smaller CPS score (less cognitively impaired) are more than twice as likely to be at least bachelor educated and 22 percentage points more likely to be responsible for making decisions about their care. Finally, over relevant ranges nursing homes exhibit economies of scale (but at a decreasing rate). For example, around 80% of nursing homes have 150 beds or less. At 150 beds, costs are still decreasing with

total beds.

Table 9: Firm-level payer-type marginal costs

	Mean	25 Pct	Median	75 Pct
Private-payer MC				
FP	43.7	11.3	40.9	74.8
NFP	30.3	5.8	51.6	93.4
Medicaid payer MC				
FP	125.1	98.7	124.4	152.4
NFP	169.4	159.0	197.8	234.6
Medicare payer MC				
FP	130.2	85.4	124.0	168.7
NFP	169.3	144.6	190.5	232.0

Note: All costs in \$/bed-day. We winsorize the estimated marginal costs by dropping estimates below the 1% quantile for the sum of the zip fixed effect and unobserved marginal cost shock. These reflect cases where the model fit is especially poor.

Next, focusing on comparisons across FP and NFP providers, we find that FP providers generally have lower marginal costs across Medicaid and Medicare payers, of around \$35-40 per bed day. To the best of our knowledge, there are no other papers which study whether FP providers have more efficient cost functions than their NFP counterparts. Lu and Lu (2021) study conversions from NFP to FP nursing homes and find that converted FP nursing homes are able to reduce costs by reducing nursing administration costs which is consistent with some FP providers having more efficient cost functions.⁴⁶ In the absence of other empirical evidence, we can apply the framework of Hart et al. (1997) who argue that government employee incentives to reduce costs are weaker than for a private firm as they do not appropriate all the gains of their efforts. Around 20% of our NFP providers are publicly owned. For the remaining 80% a similar argument applies. NFP providers are required to reinvest any profits back into the organisation, and regulation works to limit the ability of firms to distribute profits through higher wages. Thus, NFP employees will also be unable to appropriate additional profits from cost reduction.

Still these cost differences are large, thus it's worth exploring systematic patterns that could explain these. Identification of cost parameters is partly driven by the matching of variable costs, thus variation from cost report data is relevant to consider. In table 10, we study how average spending from different cost-centers reported in the cost report data vary across FP and NFP providers. In

⁴⁶The generalisability of Lu and Lu (2021)'s finding is unclear. This is because their identifying variation is based on nursing homes which convert from NFP to FP, which they suggest are often 'failing' NFP nursing homes.

each column we regress the average cost in each cost center on a NFP indicator with zip and year fixed effects. In the bottom row, we also present the relative sizes of these cost centers by showing their average relative proportions. The first column focuses on the skilled nursing cost center, which captures arising directly from the provision of skilled nursing care. We deduce the wage bill of licensed nurses (registered nurses and licensed practical nurses) as the costs of these are directly captured in the model. The most notable component in this remainder would be the wage costs of certified nursing assistants. As table 2 showed, certified nursing assistants (CNA) make up the majority of all nursing hours, and NFP providers choose more CNA nursing hours per bed-day of 2.26 versus 1.88 for FP providers. CNAs do not show up in our cost function explicitly but are a part of variable costs in the data through the skilled nursing cost center, so this is likely to be a major influence in the difference between FP and NFP costs.

Ancillary covers a range of health-related expenses that support the provision of skilled nursing care. This includes for example spending on therapists, such as physical therapy and patient supplies. NFP providers appear to spend slightly less on ancillary care compared to FP providers (column 3). Finally non-medical aggregates all non nursing care cost centers such as housekeeping and food. Essentially, this represents costs related to accommodation and living. NFP providers spend around \$9 per bed-day more than FP providers in this cost center, so this is likely to be another driver of cost differences between FP and NFP providers. These cost centers suggest that differences in unobserved quality could explain about \$18 of the cost difference between FP and NFP providers. For Medicaid-payers the gap is around \$45 so about 40% of this gap with the remaining 60% being differences in operational efficiency of the nursing homes.

Since we only observe the total spend as opposed to input choice, these cost differences could also represent inefficient contracting or usage as opposed to differences in non-medical quality. Overall, we think these cost differences likely reflect differences in non-medical quality. First, our demand estimates suggest that the residents most likely to value non-medical quality such as those with especially long stays, are at least bachelor educated and make care decisions for themselves have stronger preferences for NFP providers. Preferences for NFP providers are consistent with systematic differences in unobserved quality which are in line with the higher average spend by NFP providers. Another reason we find this plausible is that it is consistent with the NFP business model of operating upstream facilities (independent living and assisted living facilities) whose residents mostly pay a private price. Quality should be especially important for those facilities for attracting residents who have more outside options, for example staying at home. It seems reasonable to think of similar quality choices applying to the nursing home facility.

While these data suggest that CNAs and non-medical quality might be important we do not model them explicitly. In the case of CNAs, we did not want to introduce further endogenous variables and we found that pooling CNAs with registered nurses and licensed practical nurses

led to implausible quality preference estimates. For non-medical quality, the issue is that we do not observe the underlying inputs at all. Still, these factors are accounted for in costs indirectly through cost matching as we just described. On the demand side, they are captured by the preference for NFP providers which we found to be materially and statistically significant.

Table 10: Factors driving average cost differences

Cost Center	Skilled nursing minus licensed	Ancillary	Non-Medical
NFP	11.83 (1.40)	-2.69 (1.17)	9.05 (0.99)
Av. Proportion (%)	50.4	22.0	27.5

Note: Regressions include zip and year fixed effects with standard errors clustered at the facility level. These results use facility-level data from 2000-2010 in California.

Model Discussion

Identification of supply parameters relies on the inversion of first-order conditions being correct. Or, in other words, that the supply model is correct. This is especially so for the quantity-of-care motive parameter since it rationalises the difference between observed and model simulated variable costs. One potential concern is that, FP providers are typically part of a large chain. FP nursing homes in the same chain that are close enough may have smaller incentives to compete which could also cause issues for our inversion. Unfortunately, as chain ownership is opaque in the data, we have not been able to account for chain effects. In future drafts, we may complete the hand-coding of chains to enable us to do so, however, there are also reasons to suggest that chain effects are limited.

First, our in-sample takeovers presented in appendix A.5 provide some tangential evidence about this. Most of the FP to FP acquisitions that we observe are likely to be from larger chains acquiring other FP nursing homes. New owners appear to improve profits by changing the composition of the nursing home population towards Medicare payers while being able to maintain costs.⁴⁷ There do not seem to be otherwise any significant changes in price and quality. These results accord to some extent with Gupta et al. (2021) who explicitly study takeovers by private equity. While they do not look at changes in payer-type populations, they find similar results on nursing hours and costs.

⁴⁷Since our model estimates suggest that Medicare payers are the most costly, in the lens of our model, the maintenance of cost would have to be due to systematic changes to the cost function outside of the scope of the model.

In part, we expect that the limited pricing and quality effect is because nursing homes which are part of chains often still operate independently to a significant extent. For example, the largest nursing home chain in the US,⁴⁸ - the Ensign group - describes on their website a business model where each nursing home is run independently by its own management team and employees.⁴⁹ In addition, in our hand-coding exercise, it was often the case that nursing home webpages showed little about their owner or even a consistent branding for nursing homes we suspected to be of the same chain.

Another issue that is important to discuss is equilibrium existence. Profit functions are not guaranteed to be quasi-concave in our setting because the nursing home's single private price and quality also govern revenue from Medicaid and Medicare payers through demand. In addition, marginal costs are not necessarily convex as the curvature of the cost function depends on the population distribution of payer types within a market. Therefore equilibrium existence is unclear. Imposing more curvature in the cost function such as by allowing for the Medicaid and Medicare quantities to have quadratic terms could mitigate these issues. The challenge is that this introduces more parameters, requires more instruments and requires more of the data. The imprecision of some of the supply side parameters led us to avoid taking this approach.

On top of structural reasons, as we saw in figures 3 and 4 there are also likely poor fit issues for some nursing homes. In practice, this means that our estimated parameters do not always rationalise the data. Specifically, in markets (MSA-years) with no more than 20 nursing homes,⁵⁰ 12 out of 168 (7%) NFP nursing homes and 157 out of 613 (26%) FP nursing homes do not satisfy their second-order conditions. This leaves us with 43 out of 108 markets where equilibrium conditions are satisfied for every nursing home in the market.

8 Explaining Quality and Access Patterns

With our estimated model of nursing home demand and supply, we can now revisit and explain our motivating evidence through the lens of the model. We begin by conducting a series of counterfactuals to explain quality choices and market segmentation. In all counterfactuals conducted in this paper, we focus on the 43 small markets (MSA-years with less than 20 nursing homes) satisfying equilibrium conditions as previously discussed. For the counterfactuals in this section, we change the relevant supply parameters for NFP providers and then solve for the new pricing and quality equilibrium. Doing so allows us to explore which forces can explain why NFP providers

⁴⁸According to Definitive Healthcare, see <https://www.definitivehc.com/blog/top-10-largest-skilled-nursing-facilities>

⁴⁹See for example <https://ensigngroup.net/about/>

⁵⁰For counterfactuals, we restrict consideration to markets with no more than 20 nursing homes for tractability

choose higher quality than their FP counterparts

We consider changes to three parameters. First, since we found that NFP providers have a non-pecuniary motive for quantity-of-care, it could be the case that this causes them to choose higher quality to serve more residents. Second, NFP providers may prefer to avoid directly competing with FP providers and their cost advantage by serving residents with stronger preferences for quality. Thus differences in marginal costs could also lead to differences in quality choices. Finally, changes to marginal costs hold fixed incentives for targeting service to particular payer types. Since Medicaid bed-days make up the majority of bed-days served on average, changes in Medicaid quantity costs are especially strong in changing incentives to target service to particular payer types and could also have effects on quality choices.

Figure 5 presents the results from these counterfactuals. In each plot, we present the average price and quality choice across FP and NFP providers in all 43 markets under 7 different choices of supply parameters. In the first three plots we consider the baseline (estimated parameters) and reductions in the stated parameter by 5, 10, 15, 20, 25 and 30. We add arrows to indicate the direction in which average prices and quality choices change as the magnitude of the change in supply parameters changes. In the top left figure, we study how price and quality choices adjust as the NFP quantity motive is reduced. Given our quantity motive estimate for NFP is 28.34, this series of counterfactuals essentially removes the NFP non-pecuniary motive. Weakening this motive leads NFP providers to derive less utility from serving more beds, so they increase their prices and reduce quality to reduce demand for their nursing homes. The extent to which the adjustment occurs through price or quality is ambiguous and depends on the distribution of preferences for price and quality. In fact, while there is substantive adjustment on both dimensions in reaching a counterfactual with no non-pecuniary motive, the steps are not even. For example, the initial response is to mostly reduce quality with minor increases in price. NFP providers tend to serve residents with stronger preferences for quality, thus reducing quality first has a larger impact on reducing quantity of care served. While NFP providers become less competitive, especially for consumers with stronger preferences for quality, FP providers are able to increase prices and quality, further widening the gap in quality choices. Either way, these counterfactuals suggest that quality differences are partly explained by the difference in non-pecuniary motives.

Reducing NFP marginal costs by 30 eliminates the average marginal cost gaps on serving Medicaid payers, almost eliminates the average marginal cost gap for Medicare payers and widens the average gap and cost advantage that NFP providers have in serving private payers. Accordingly, NFP providers significantly reduce their prices. Their responses to quality vary significantly with the size of the reduction in marginal cost. Initially, quality declines. As NFP providers close the cost gap, their price reductions allow them to compete more effectively with FP providers and for residents with weaker preferences for quality. Accordingly the value of higher quality declines

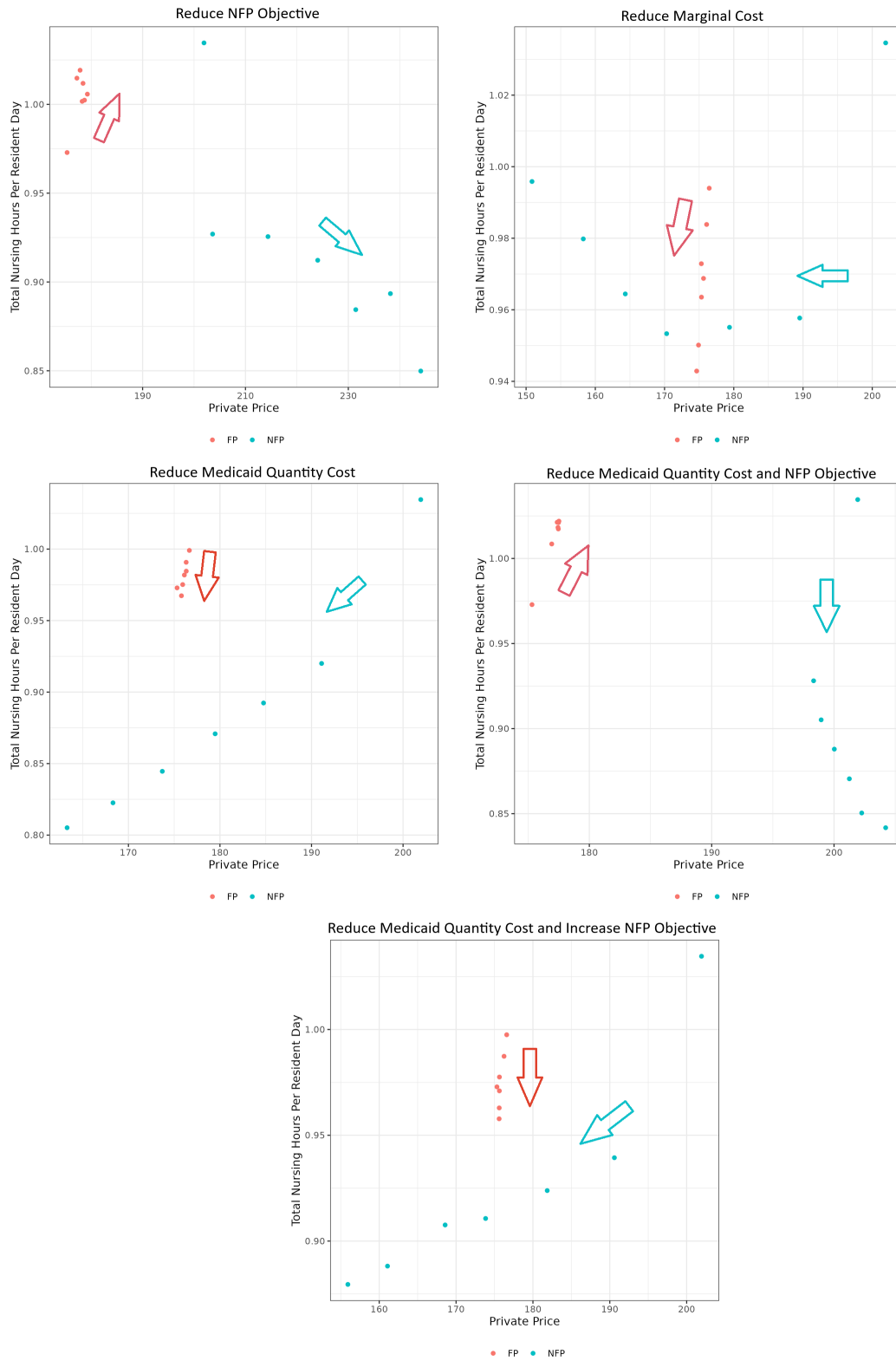


Figure 5: Counterfactuals to explain quality choices

and they reduce their quality. As NFP prices fall below the prices of FP providers, marginal revenues from further decreases in price decline as NFP providers have already captured much of the share of residents with weaker preferences for quality. At this point, NFP providers can offer a superior product compared to FP providers by raising quality. The main response by FP providers is to reduce quality to protect their margins. This is consistent with demand estimates that suggest that FP providers were already serving residents with weaker preferences for quality. Given the ambiguous responses to quality, it's unclear from these counterfactuals alone whether quality differences can be attributed to differences in costs.

Next, counterfactuals studying reductions in Medicaid quantity costs highlight how differences across payer types drive pricing and quality choices. While reductions in marginal costs resulted in ambiguous responses to quality, now NFP providers reduce quality and prices as the gap in Medicaid quantity costs is almost eliminated. Since Medicaid payers have weaker preferences for quality, NFP providers find it more valuable to continue reducing quality instead of eventually raising quality. FP providers again primarily adjust by reducing quality but proportionally less than for NFP. Thus differences in Medicaid quantity costs specifically could cause differences in quality choices.

In summary, differences in quantity motives and costs have the potential to explain differences in quality choices. Therefore, to narrow down which forces, if any, really matter, we need to consider counterfactuals where we adjust multiple parameters at once. First, given the differences in estimated parameters and marginal costs across FP and NFP, it is reasonable to restrict attention to the quantity motive and Medicaid quantity costs. A reduction in marginal costs of \$30 would exacerbate the cost advantage that NFP providers have over FP providers for private payers, not necessarily making them more alike. In the last two plots we consider changes to both Medicaid quantity costs and quantity motives by 2.5, 5, 7.5, 10, 12.5 and 15. Following the results from unilateral policy changes, reducing both the quantity motive and Medicaid quantity reduces the quality of NFP providers. Changes in price are more informative. NFP prices initially decline, consistent with the declines in price from reduced Medicaid quantity costs. Eventually, they rise, as the reduced quantity motive matters more. These counterfactuals suggest that both NFP quantity motive and higher cost matter for their quality and pricing choices.

Since reducing quantity motives and Medicaid quantity costs for NFP providers both tend to reduce their quality choices, one other exercise that is worth asking is which one matters more for determining quality choices? Our last series of counterfactuals considers this by reducing Medicaid quantity costs and increasing the quantity motive instead of reducing it. Reducing quality all else equal reduces demand for the nursing home, thus increasing quantity motives could lead to higher quality choices. Our counterfactuals demonstrate that the reduction in Medicaid quantity costs dominate. NFP providers reduce quality to enable larger reductions in price. Thus one

interpretation is that overall differences in cost functions matter more than quantity motives for determining why NFP providers choose higher quality than their FP counterparts.

Finally, we can turn to explaining patterns of access. Our motivating evidence established that NFP providers are more prevalent in higher income markets and serve less Medicaid and more private-pay residents than their FP counterparts. Since nursing homes are able to serve more private payers in higher income markets and more Medicaid residents in lower income markets, we hypothesised that the patterns of access can be explained by FP providers finding Medicaid residents more profitable and NFP providers finding private-pay residents more profitable. Since payer-type margins are not directly observed, this was one of the motivations for the structural model.

Focusing on figure 4, the rightwards shift in the distribution of Medicaid margins for FP relative to NFP providers is a clear indication that FP providers find Medicaid residents profitable compared to NFP providers. On the NFP side, Medicaid margins appear to be the worst on average with the private margin being largest. This is consistent with the observed patterns of access. The fact that a large proportion of Medicaid margins is negative is also consistent with industry reporting. For example, the American Health Care Association reports that two-thirds of surveyed nursing homes are reimbursed for less than 80% of the cost of providing care for Medicaid payers and only 10% report reimbursements which cover 90-100% of the cost of providing care for Medicaid payers (American Health Care Association, 2025).

9 Banning for-profit providers

Finally, we use the estimated structural model to simulate the effects of counterfactual bans, allowing for subsequent takeovers by NFP providers. In each market, we consider a range of scenarios, beginning with the lowest quality FP nursing home and cumulatively targeting higher quality FP nursing homes, ending with the scenario where half of FP nursing homes are targeted.

In a 'takeover', the targeted FP nursing home is taken over by a local 'NFP' nursing home. Since our pool of NFP nursing homes includes government providers (around 20% of NFP providers), we could also think of this as government directly taking over the nursing home. The takeover aspect of the counterfactual is important. A counterfactual based only on bans is almost certainly going to reduce consumer surplus and welfare. Removing nursing homes means that some residents may lose their most preferred nursing home, while others may lose access to nursing home care altogether, leading to large losses in consumer surplus and firm profits and utility.⁵¹ On

⁵¹Counterfactual simulations suggest that even in scenarios where only the worst FP provider is banned the average and median loss in consumer surplus is \$27.4 million and \$26.6 million respectively, while the average and median loss in total bed-days served is 30,878 and 31,515 respectively.

the other hand, the effect of takeovers is ambiguous. If NFP providers' quantity motive is strong enough, replacing FP providers with NFP providers could intensify competition and improve quality. While in most papers, the relevant extensive margin response is new entry, as we discussed in the motivating evidence, new nursing home entry seems relatively unlikely and takeover seems like a more reasonable proposition.

In terms of implementation, for the ban counterfactual, we remove the targeted FP providers and solve for the new equilibrium. In the takeover counterfactual, we change parameters on the demand and supply side for the targeted nursing homes and then solve for the new equilibrium. On the demand side, we add the NFP 'product attribute' to targeted nursing homes, so that residents take into account their preference for NFP providers when evaluating the nursing homes which are taken over. On the supply side, we change all of the FP parameters with their NFP counterparts. We replace the unobserved cost shocks with the mean of the corresponding NFP cost shocks.

These changes represent a counterfactual where a local NFP nursing home replaces the FP 'product' with their own. While total nursing hours remains endogenous, the NFP taking over the FP introduces higher unobserved quality in line with its existing product offering and incurs the associated higher level of cost. This is a reasonable counterfactual to consider to the extent that it seems natural that the NFP would partially extend its existing business model given the 'local' nature of the takeover. We do not explicitly model a takeover process and any transition costs incurred. To that extent, we regard our evaluation as generous to the policy. Given the static nature of our model and counterfactual, we also interpret the results from a long-run point of view.

Table 11: Aggregate Market Effects from Takeovers

	Mean	25th Pct	Median	75th Pct
CS (\$m)	-3.71	-7.20	-3.34	-0.81
Total Quantity (bed-days)	-270	-87	-14	0

Note: This table summarizes market-level impacts of counterfactual bans and takeovers. Each unit corresponds to a scenario - one market with one set of targeted FP providers. The sets of targeted FP providers for a given market consist of a set with the lowest quality FP provider, a set with the lowest quality and second lowest quality FP provider, and so on, with the largest set half the number of FP providers (rounded up). Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. Since costs are heterogeneous, the changes need not be monotonic in the number of targeted FP providers.

Table 12: Firm Level Effects from Takeovers

		Mean	25th Pct	Median	75th Pct
Price	Remaining FP	3.55	-12.02	-0.95	9.36
	Takeover NFP	90.73	18.32	68.52	128.96
	Existing NFP	14.78	-6.32	0.11	19.39
Quality	Remaining FP	-0.003	-0.094	-0.006	0.038
	Takeover NFP	-0.001	-0.514	-0.146	0.120
	Existing NFP	0.078	-0.048	0.007	0.051

Note: This table summarizes the firm-level impacts of counterfactual bans and takeovers. Each unit corresponds to a group of providers in a scenario - one market with one set of targeted FP providers. The sets of targeted FP providers for a given market consist of a set with the lowest quality FP provider, a set with the lowest quality and second lowest quality FP provider, and so on, with the largest set half the number of FP providers (rounded up). Changes are computed relative to the observed differentiated Nash-Bertrand equilibria. Since costs are heterogeneous, the changes need not be monotonic in the number of targeted FP providers.

Tables 11 and 12 present the results of these counterfactuals. Overall bans, even allowing for takeovers cause considerable resident harm. Consumer surplus reduces in the majority of scenarios with reductions of around \$3.3 million and \$3.7 million in the median and mean scenario.⁵² There are minor reductions in the number of bed-days served, thus most of the reduction in consumer surplus is arising from degraded services which the firm-level effects table unpacks.

First, focusing on the direct impact, NFP providers that takeover a FP provider significantly raise private prices. In the median and mean scenario private prices are raised by around \$70 and \$90 respectively. Part of the increase in private price reflects existing differences in private prices between FP and NFP providers in the observed equilibria of around \$15 and \$25 in the median and mean respectively. The remaining \$55 or \$65 is the NFP provider requiring cross-subsidization from private payers for serving Medicaid payers. This large increase is consistent with the patterns in the data and other structural estimates. First, the average NFP provider has a negative margin on serving medicaid payers. Second, our demand estimates suggest that residents exhibit significant inertia. Thus even under these substantive counterfactuals, many residents would be unlikely to change nursing home. The NFP provider taking over the FP provider would have to serve a larger proportion of Medicaid residents which their usual business model, with negative margins on Medicaid payers is unsuited for. This requires larger cross-subsidization by private

⁵²Consumer surplus is calculated as $CS = \sum_i Days_i CS_i$, where CS_i is the usual logit logsum formula for consumer surplus. When expressing consumer surplus in dollars, we assume that residents who never pay the private price otherwise have the same price preferences as with someone who does.

payers, which due to inertia is feasible. The response on the quality margin is ambiguous. In many cases, following takeover, the NFP provider reduces quality to reduce costs as indicated by the negative median change of 0.146, while in others the NFP provider increases quality, for example the third quartile change is 0.120. The average difference in quality between FP and NFP providers is 0.062, thus these changes can be large as well.

Finally, we turn to the indirect impacts. Overall, the indirect impacts are smaller. The magnitude of changes is much smaller, compared to the providers being taken over and there is more ambiguity in the direction of change. In the average and median scenario, the remaining FP providers shade down their quality as the provider being taken over, often reduces their quality (which by construction have the lowest quality already). Accordingly the median private price for the remaining FP providers also declines. Existing NFP providers increase their quality in the average scenario. This reflects that some providers that take over FP providers significantly increase quality, resulting in the need for existing NFP providers to compete on quality.

10 Conclusion

Should policymakers reduce FP provision and increase NFP provision in regulated markets such as health and education? Overall, we argue that the balance of the model and empirical evidence suggests that it does not. In some other settings such as hospitals, papers have argued that NFP providers may simply be FP providers in disguise (Dranove et al., 2017). We find significant differences between NFP and FP providers. Consistent with past work, NFP providers have non-pecuniary motives for providing quantity of care. However, unlike in previous papers, we argue that this motive plays a relatively small role in determining market structure. NFP providers also choose higher quality and likely do so on more dimensions than that which we explicitly model. However, higher quality comes at a ‘cost’. Furthermore, as Medicaid payers – who account for the majority of bed-days – value quality less, differential costs ultimately become a first-order consideration in determining market structure and nursing home choices. Viewing FP providers as offering a product better suited and providing access to Medicaid payers, it is clear that they play an important role in nursing home care and that banning FP nursing homes is unlikely to be a fruitful policy solution.

A broader takeaway is that given the limited policy success in this industry, one might have hoped that NFP provision could be a silver bullet. This is especially the case because it does not necessarily involve the government spending substantially more money, which history demonstrates is a difficult ask. Ultimately, these results suggest that increased NFP provision is unlikely to be a substitute for increased government funding through measures such as paying nursing

homes more as in Hackmann (2019) or a stronger regulator, which was a key recommendation in National Academies of Sciences, Engineering, and Medicine (2022). Finally, we began this paper with general health and education settings in mind. We leveraged the US nursing homes industry to study this question, which necessitates addressing unique institutional features. However, similar patterns can be found in other markets such as post-secondary education (Deming et al., 2012). Accordingly, we contend that this framework and the lessons learned may have broader applicability beyond the US nursing homes industry.

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Appendix

A Supplemental facts

A.1 Resident individual characteristics

Table 13: Summary statistics from resident-level data

	Average	10th pct	50th pct	90th pct
ADL score	12.5	7	13	17
Age (cond. ≥ 65)	82	71	82	92
CLN score	1.3	0	1	2
Length of Stay (days)	148	7	30	410

This data is from the resident-level panel which informs our structural estimation and covers 2000-2010 in CA.

According to table 13, residents vary significantly in their length of stay, entering age of residents,⁵³ care needs and medical needs. Care needs can be measured by a resident's assisted daily living (ADL) score which measures how dependent residents are on nursing care for their daily living. Scores can range from 4 to 18, with higher scores indicating more dependency. 4 can be thought of as normal functioning, while 18 would be someone who is completely dependent on nursing care. Medical needs can be summarised by a resident's clinically complex scale (CLN) score which takes values of 0,1,2 or 3, with 0 representing no additional medical needs and 3 representing a high level of medical needs.

⁵³We restrict the age of residents to those at least 65 to exclude residents who are likely to be very different from the population of primary interest. The long-term care function means that nursing homes may also serve young people with significant disabilities, for example.

A.2 NFP nursing homes charge higher private prices than FP nursing homes in California

Table 14: Differences in Private Prices

	Difference in Means	Regression
FP	-17.8	-15.7 (4.92)

Note: These results use yearly facility-level data from 2000-2010 in California. Controls include facility size (in beds), aggregate assisted daily living score which measures how dependent the population is on carer support for daily needs, aggregate case mix index which measures how much medical care the population requires and proportion of Medicaid and Medicare payers in the facility. Year and zip fixed effects are included with standard errors clustered at the facility-level.

A.3 Donations are unlikely to play a role in explaining location patterns

We argue that one reason why NFP providers are relatively more prevalent in higher income markets is because they serve a larger proportion of relatively profitable private-pay residents. Another reason could be that it is easier to source donations when the local population is wealthier on average. In interviews with industry, we heard that donations do not play much of a role as the cause of nursing home care is relatively unappealing. We check this claim empirically by drawing on supplemental data from the National Center for Charitable Statistics (NCCS) who compile and clean data from NFP annual form 990 filings. This data reports the contributions which nursing homes receive each year along with usual financial data. Contributions includes both donations and government funding, but are not separately split in the NCCS data. Figure 6 is a histogram of nursing home contribution rates as a proportion of total revenues separated by the income quartile of the nursing home's zip. A few nursing homes receive all of their revenue in the form of contributions. These are likely to be government-run nursing homes. On the other hand, the vast majority receive none, and zip income quartile has very little effect on this distribution. We interpret this as clear evidence that donations do not matter.

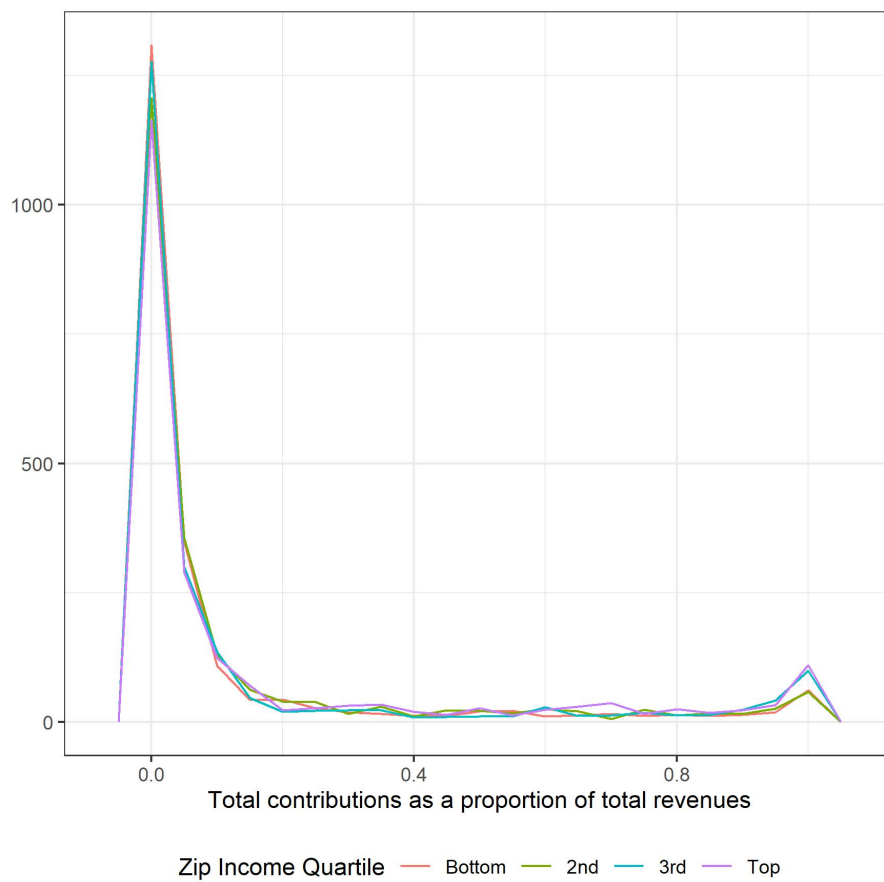


Figure 6: Histogram of contribution rates by zip income quartile, all states, 2010-2018

A.4 Nursing composition correlates with the distribution of payer types served

Table 15: How the proportion of LPNs varies with payer types served

	(1)	(2)
Prop Mcaid	0.059 (0.025)	0.004 (0.033)
Prop Mcare	-0.082 (0.059)	-0.104 (0.057)
Fixed Effects	Zip	Facility

Note: These results use yearly facility-level data from 2000-2010 in California. In addition we control for total beds and year fixed effects. Standard errors clustered at the facility-level.

Here, we look to explain variation in licensed practical nursing (LPNs) hours as a proportion of total licensed nursing hours (the other type being the higher-skilled registered nurses). In column (1) we examine variation across facilities and observe that a larger proportion of Medicaid payers being served correlates with more LPN usage. In column (2) we examine variation within facilities and observe that a larger proportion of Medicare payers being served correlates with less LPN usage.

A.5 Validating in-sample nursing home takeovers

Motivated by papers studying acquisitions and the lack of exit and entry activity in the data, we consider takeover activity in CA in table 5. As facility data do not readily identify ownership, and thus takeovers, we had to hand code takeovers by cross-validating the facility-level data with online sources. To provide some reassurance that we did successfully identify takeovers we conduct an event-study exercise.

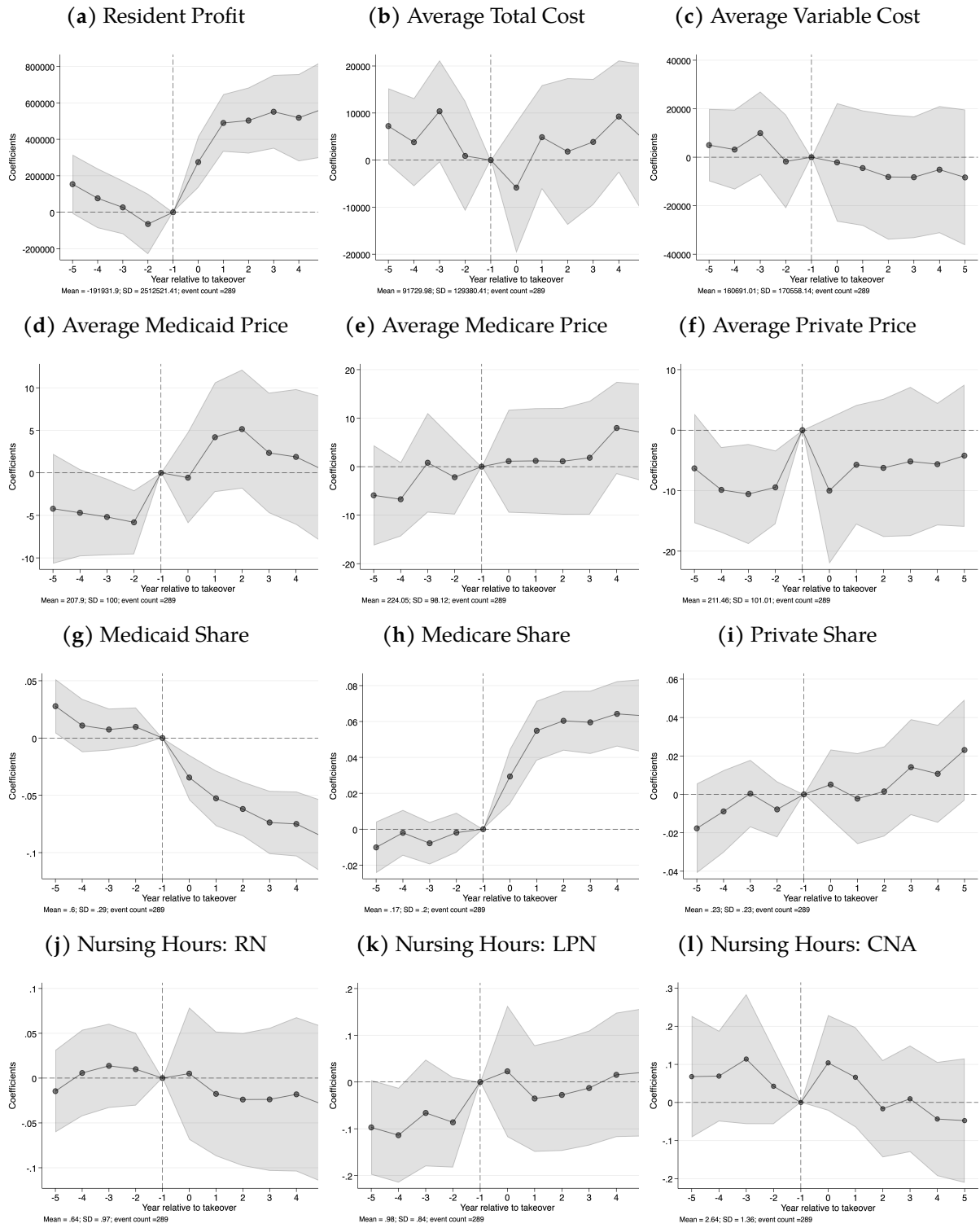
Specifically, we estimate the following regression equation for facility j at year t :

$$y_{jt} = \alpha_j + \tau_t + \sum_{k \neq -1} \beta_k \mathbf{1}_{j,k=r(j,t)} + \epsilon_{jt}$$

where α_j are facility fixed effects, τ_t are year fixed effects, and $r(j, t)$ is the relative year from when the facility j was taken over by another facility. Figure 7 reports the estimated coefficients β_k .

The sample is relatively small so our estimates are somewhat noisy. Some clear patterns emerge. resident profit which are revenues minus expenses for resident care items increases substantially following a takeover. The reason appears to be that nursing homes shift away from serving Medicaid residents towards serving other resident types while keeping average variable and total costs in check. Since, Medicare and private pay generally pay more, if a nursing home can keep costs fixed when shifting towards Medicare and private pay residents then their resident profitability will increase. Notably, nursing hours and the prices themselves do not significantly change. The result on unchanging nursing hours seems close enough to Gupta et al. (2021) whose results suggest no significant change in nursing hours until much later after the takeover. They do not study changes in prices or the payer-type composition. They also find significant increases in facility fixed costs following a takeover, which in our view is likely attributable to the tunnelling of profits which Gandhi and Olenski (2024) find. Our event study also suggests that fixed costs rise following takeovers, but we do not have enough power to conclude statistical significance.

Figure 7: Event-study coefficients: Takeover



Nursing hours are hours per resident per day.

B Supplements to structural estimation

B.1 Full demand parameter estimates

Table 16: Full demand parameter estimates

	Baseline	Age (10 yrs)	Los 30-100	Los 100+	Resp. Non-self	Payment Self	Payment Medicaid	Payment Medicare
Distance	-1.9950	-0.2706	-0.1165	-0.1486	-0.1198			
(10 miles)	(0.0069)	(0.0107)	(0.0022)	(0.0041)	(0.0139)			
Distance Sq	-0.0138	0.0500	0.0507	0.1361	0.0833			
(10 miles sq)	(0.0068)	(0.0048)	(0.0036)	(0.0041)	(0.0080)			
Quality	-1.3256	0.2346	-0.1823	-0.2966	-0.7458		-0.3370	0.6284
	(0.7469)	(0.0025)	(0.0006)	(0.0016)	(0.0081)		(0.0050)	(0.0033)
Price	-0.0248	0.0018	0.0009	-0.0001	-0.0001	-0.0017		
	(0.0051)	(0.00002)	(0.0001)	(0.0001)	(0.0001)	(0.0002)		
Inertia	3.9590							
	(0.0029)							
NFP	-4.0118	0.5142	-0.2035	0.3653	-0.2146		-0.5269	0.8266
	(0.2177)	(0.0007)	(0.0010)	(0.0035)	(0.0011)		(0.0086)	(0.0039)
	DRS	CPS	ADL	CLN	Has Alz.	Female	Non-white	Bach. Educ.
Quality	-0.0166	-0.0669	0.0785	0.2321	-0.3193	0.0265	-0.1729	0.3231
	(0.0025)	(0.0012)	(0.0007)	(0.0004)	(0.0004)	(0.0019)	(0.0038)	(0.0037)
Price								0.0011
								(0.0002)
NFP	0.1052	-0.0115	-0.0876	0.0230	-0.0510	0.1801	-0.4103	0.6574
	(0.0003)	(0.0002)	(0.0002)	(0.0001)	(0.0012)	(0.0001)	(0.0020)	(0.0020)

Note: These results use individual-level data from 2000-2010 in California. Standard errors were calculated by block bootstrap (drawing nursing homes) for 400 iterations.